HYBRID CONVOLUTIONAL NEURAL NETWORK-VISION TRANSFORMER ARCHITECTURE WITH PARALLEL FEATURE PROCESSING AND CROSS-MODAL ATTENTION FOR LUNG SEGMENTATION

by

Khoa Pham Nguyen Thanh

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Professional Master in Data Science and Artificial Intelligence Applications

Examination Committee:	Dr. Chutiporn Anutariya (Chairperson)
	Dr. Cherdsak Kingkan (Co-Chairperson)
	Dr. Chaklam Silpasuwanchai
	Dr. Chantri Polprasert
Nationality:	Vietnamese
Previous Degree:	Bachelor of Science in Geology
	University of Science
	Viet Nam National University Ho Chi Minh City

Asian Institute of Technology
School of Engineering and Technology
Thailand
December 2025

AUTHOR'S DECLARATION

I, PHAM NGUYEN THANH KHOA, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

Date: November 15, 2025

Name:

Khoa Pham

Signature:

ACKNOWLEDGEMENTS

I feel incredibly grateful for my family, who have been my strong back through this entire journey. They have stood by me through all the ups and downs, supporting me as I needed to concentrate on studying and working on projects, assignments and exams and always believing in me when things got challenging. I also feel deeply thankful for all the professors in the Data Science and AI program at AIT. They have taught me so much and inspired me to think in new ways. Their guidance and support have meant everything to me as I have grown both as a student and as a person. This whole experience has opened my eyes to new possibilities and shown me what I can achieve. I know this time in my life will be something I look back on as a real turning point—one that has shaped not just what I know, but who I am and who I want to become.

ABSTRACT

Chest X-ray lung segmentation is essential for automated diagnosis of respiratory diseases, yet current approaches using either CNNs or Vision Transformers independently often is not fully captured all context of chest X-ray images . These images require both precise local boundary detection and global anatomical understanding due to overlapping structures and projection artifacts. While hybrid CNN-Transformer models have succeeded in computer vision, their application to medical image segmentation remains largely unexplored. This thesis presents hybrid architecture combining CNN and Transformer approaches for chest X-ray lung segmentation. The model employs parallel processing pathways where a CNN branch extracts local features and a Transformer branch captures global context. The key is a cross-modal attention mechanism enabling information exchange between branches. Through multi-head cross-attention, the CNN learns global context from the Transformer while the Transformer acquires spatial details from the CNN, creating enhanced representations that leverage both local precision and global coherence. The fused features are processed through a decoder with skip connections. Evaluation on standard datasets demonstrates improved segmentation performance compared to existing methods while maintaining computational efficiency. This research contributes the a hybrid CNN-Transformer architecture specifically designed for chest X-ray lung segmentation, establishing a foundation for future medical hybrid models.

Key words : Chest X-ray segmentation, Hybrid CNN-Transformer, Cross-modal attention, Lung segmentation, Medical image segmentation, Vision Transformers, Feature fusion, Automated diagnosis

CONTENTS

	Page
ACKNOWLEDGEMENTS	**iii**
ABSTRACT	**iv**
LIST OF TABLES	**viii**
LIST OF FIGURES	**ix**
CHAPTER 1 INTRODUCTION	**1**
1.1 Overview	1
1.2 Problem Statement	2
1.3 Objectives	3
1.4 Limitations and Scope	4
1.5 Thesis Outline	4
CHAPTER 2 LITERATURE REVIEW	**6**
2.1 Traditional Segmentation Methods	6
2.2 CNN-Based Approaches	7
2.2.1 U-Net and Its Variants	8
2.2.2 Modern CNN Architectures	8
2.3 Vision Transformers in Medical Imaging	9
2.3.1 Transformer Fundamentals	10
2.3.2 Medical Imaging Applications	11
2.3.3 Challenges and Limitations	12
2.4 Hybrid CNN-Transformer Architectures	12
2.4.1 Pioneering Hybrid Models	12
2.4.2 Fusion Strategies	14
2.4.3 Feature Fusion Mechanisms	14
2.4.4 Attention Mechanism Integration in Hybrid Architectures	16
2.4.5 Application Gaps in Chest X-ray Segmentation	17
2.5 Recent Advances and State-of-the-Art Methods	17
2.5.1 SAM-based Approaches	17
2.6 Current Benchmarks and Evaluation Methods	18
2.6.1 Standard Datasets	18

2.6.2 Evaluation Metrics 18
2.6.3 Loss Functions 21
2.6.4 Performance Evaluation Considerations 23
CHAPTER 3 METHODOLOGY 24
3.1 Overview 24
3.2 Dataset Introduction 24
3.2.1 Data Sources and Composition 24
3.2.2 Technical Image Specifications 27
3.2.3 Dataset Partitioning and Validation Strategy 27
3.2.4 Dataset Quality Assurance and Limitations 28
3.3 Labeling Method 28
3.3.1 Ground Truth Annotation Process 28
3.3.2 Annotation Output 28
3.4 Proposed Hybrid Model Architecture 29
3.4.1 Architectural Design 29
3.4.2 CNN Pathway Design 30
3.4.3 Vision Transformer Pathway 31
3.4.4 Cross-Attention Fusion Strategy 32
3.5 Decoder Architecture and Output Generation 32
3.5.1 SimpleDecoder Design 32
3.5.2 Complete Architecture Flow 33
3.5.3 Addressing Current Limitations 34
3.5.4 Advantages of the Proposed Approach 34
3.5.5 Potential Limitations and Considerations 34
3.6 Loss Function Design 35
3.6.1 Multi-Component Loss Strategy 35
3.6.2 Component Loss Functions 35
3.6.3 Dynamic Weighting Strategy 36
3.7 Evaluation Metrics 36
3.7.1 Overlap-Based Metrics 36
3.7.2 Boundary-Specific Metrics 37
3.7.3 Clinical Relevance Metrics 37
3.8 Experimental Design and Validation 38

3.8.1 Training Protocol 38
3.8.2 Ablation Studies 38
3.8.3 Baseline Comparisons 38
CHAPTER 4 EXPERIMENT 40
4.1 Experiment Configurations 40
4.1.1 Training Scenarios 40
4.1.2 Architectural Variants 41
4.2 Experiment Results 41
4.2.1 Scenario 01: Baseline Performance on MC+SHZ Dataset 41
4.2.2 Scenario 02: Cross-Dataset Generalization to Qatar-COVID-19 47
4.2.3 Scenario 03: Enhanced Training with Diverse Data 54
4.2.4 Comparative Analysis with State-of-the-Art Methods 60
4.2.5 Ablation Study Analysis 63
4.2.6 Model Complexity and Performance Trade-offs 67
CHAPTER 5 CONCLUSION & DISCUSSION 69
REFERENCES 72

LIST OF TABLES

Tables **Page**

Table 4.1 Performance comparison for models trained and tested on the MC+SHZ dataset. Best results are highlighted in bold. 46

Table 4.2 Cross-dataset generalization performance. Models were trained on MC+SHZ and tested on the unseen Qatar-COVID-19 dataset. 53

Table 4.3 Performance on the Qatar-COVID-19 test set after enhanced training with MC+SHZ and additional normal COVID-19 images. 59

Table 4.4 Comparison with state-of-the-art methods. Dataset abbreviations: M=MC, S=SHZ, C=COVID19, P=Private, Q=QaTa-COV19. Results for FAT-Net are from Khorasani and Mofrad Khorasani and Mofrad (2025). 61

Table 4.5 Performance comparison on the Shenzhen dataset. Benchmark results for U-Net, UNet++, TransUNet, and Swin-Unet are adapted from Yao et al. Yao et al. (2023). 62

Table 4.6 Performance comparison of the full hybrid model and its architectural ablations (CNN-Only, ViT-Only) across three experimental scenarios. The 'Full Hybrid' model serves as the baseline, with ΔDice and ΔHD95 metrics calculated relative to its performance in each scenario. Scenario 1 (Standard Training) tests on the MC+SHZ dataset. Scenarios 2 (Cross-Dataset Generalization) and 3 (Expanded Training) test on the unseen Qatar-COVID-19 dataset. 65

Table 4.7 Performance comparison of different loss configurations for the hybrid model in Scenario 3. All Δ metrics are relative to the 'Unweighted Losses' baseline. Bold values indicate improvements over baseline; regular font indicates degradation. Note: For HD and HD95, negative Δ values represent improvements (lower is better). 66

Table 4.8 Model Architecture Performance and Efficiency Trade-offs 68

LIST OF FIGURES

Figures **Page**

Figure 2.1 Architecture of Deep Encoder-Decoder Convolutional Neural Networks -Kalinovsky and Kovalev (2016) 7

Figure 2.2 Unet architecture overview 9

Figure 2.3 Vision Transformer architecture overview 10

Figure 2.4 TransUnet architecture overview 13

Figure 2.5 TransFuse architecture overview 13

Figure 2.6 Fusion Strategy taxonomy 15

Figure 3.1 Methodology Diagram 24

Figure 3.2 MCU samples images & ground truth masks 25

Figure 3.3 SHZ samples images & ground truth masks 26

Figure 3.4 Qatar-COVID19 samples images & ground truth masks 26

Figure 3.5 Images and Masks folder setup 28

Figure 3.6 Labeling Process in Label Studio 29

Figure 3.7 Proposed Hybrid Model Architecture 29

Figure 3.8 Training Method 38

CHAPTER 1
INTRODUCTION

1.1 Overview

Medical image analysis has become fundamental to modern healthcare, enabling automated detection and quantification of anatomical structures across various imaging modalities. Chest radiography, constituting approximately one-third of all radiological procedures Islam and Zhang (2018), represents one of the most widespread yet challenging domains for computer vision applications.

Chest X-ray lung segmentation represents one of most challenge in computer vision task in medical image analysis—what appears straightforward to the human eye becomes remarkably difficult for automated systems. The task involves accurately capturing lung boundaries from surrounding anatomical structures, creating precise segmentation masks that can serve as the foundation for computer-aided diagnosis (CAD) systems. Within chest imaging analysis, lung segmentation serves as the cornerstone for downstream diagnostic applications. Accurate delineation of lung boundaries enables quantitative assessment of lung volume changes, monitoring disease progression over time, and provides essential regions of interest for detecting respiratory pathologies such as pneumonia, tuberculosis, and lung cancer. The precision of lung segmentation directly impacts the reliability of computer-aided diagnosis systems, making it a critical prerequisite rather than an isolated task.Shamshad et al. (2023).

Chest X-rays pose distinct challenges that set them apart from other types of medical imaging. The complexity isn't straightforward - chest radiographs have specific characteristics that make them more difficult to interpret compared to other imaging techniques used in medicine. Unlike CT scans with their clear tissue boundaries, chest X-rays project three-dimensional structures onto two-dimensional planes, creating overlapping anatomical features that cause a certain level of for many algorithms. Rib shadows cross lung fields, the cardiac silhouette obscures portions of the left lung, and pathological conditions like pleural effusions or consolidations can dramatically alter expected lung appearances Khalil et al. (2024). These complexities may explain why developing robust automated solutions for clinical deployment continues to challenge researchers worldwide.

The evolution of lung segmentation approaches has progressed from traditional rule-based methods to sophisticated deep learning architectures. Early approaches relied heavily on manual feature engineering and domain expertise, while modern methods leverage data-driven learning to automatically extract relevant features. The introduction of Convolutional Neural Networks (CNNs), particularly the U-Net architecture Ronneberger et al. (2015), fundamentally transformed the landscape by achieving unprecedented precision through automatic feature learning.

More recently, Vision Transformers (ViTs) have emerged from natural language processing, bringing remarkable capabilities for modeling global relationships through self-attention mechanisms. In medical imaging, ViTs have shown impressive results, and recent work has achieved a high benchmark for lung segmentation Shamshad et al. (2023). However, while hybrid CNN-Transformer architectures have shown remarkable success in classification tasks, their application to chest X-ray lung segmentation remains surprisingly limited. Most existing hybrid approaches focus on disease classification rather than precise boundary delineation, creating a significant research gap for segmentation-specific innovations.

1.2 Problem Statement

Current chest radiographic lung segmentation methods face several critical limitations that hinder their effectiveness in clinical applications. Traditional CNN-based approaches, while achieving high accuracy on benchmark datasets, primarily focus on local feature extraction and may miss important global contextual relationships that span large image regions. These methods excel at detecting fine-grained details, such as lung boundaries and texture patterns, but often struggle to understand the overall anatomical structure and spatial relationships between different chest components.

Conversely, pure Vision Transformer approaches demonstrate superior global context modeling capabilities through self-attention mechanisms, allowing them to understand long-range dependencies and overall image structure. However, these models often struggle with fine-grained spatial details that are crucial for precise boundary representation in medical segmentation tasks. The computational requirements of pure ViT models also present challenges for practical clinical deployment.

The fundamental problem lies in the fact that chest X-ray lung segmentation requires

both capabilities simultaneously: fine-grained local feature detection for precise boundary identification and global contextual understanding for anatomical coherence. Current methods predominantly rely on either pure CNN architectures or pure Transformer models, failing to effectively leverage the complementary strengths of both approaches.

Furthermore, most of the existing hybrid CNN-Transformer innovations in chest imaging have focused on classification tasks and other body parts segmentation such as brain tumors, multi-organ segmentation, polyp segmentation, and skin lesions - not specifically lung segmentation. This creates a significant gap in the literature, as lung segmentation tasks have unique requirements that differ substantially from classification and other body part problems. The projection-based nature of chest radiographs, with overlapping anatomical structures and variable pathological presentations, requires specialized architectural considerations that are not adequately addressed by existing general-purpose hybrid models.

1.3 Objectives

This research aims to address the identified limitations through the development and evaluation of hybrid CNN-Transformer architectures specifically tailored for chest X-ray lung segmentation. The primary objectives include:

Primary Objective: To develop a hybrid CNN-Transformer architecture that effectively combines the local feature extraction capabilities of CNNs with the global context modeling strengths of Vision Transformers to improve chest X-ray lung segmentation accuracy.

Specific Objectives:

1. **Architecture Design:** Design and implement hybrid CNN-Transformer architectures that leverage the complementary strengths of both architectures, specifically addressing the unique challenges of chest X-ray lung segmentation.
2. **Cross-Modal Attention Fusion Mechanism Development:** Develop effective fusion mechanisms that preserve fine-grained edge information from CNN components while maintaining global anatomical coherence from Transformer components, ensuring optimal integration of features.
3. **Performance Evaluation:** Conduct comprehensive evaluation of the proposed hybrid approach against existing state-of-the-art methods on standard benchmark

metrics.

1.4 Limitations and Scope

This research acknowledges several limitations that define the scope and boundaries of the investigation:

Dataset Limitations: The study primarily focuses on publicly available chest X-ray datasets including, Montgomery County, Shenzhen and Qatar-COVID19 datasets. While these provide standardized evaluation benchmarks, they may not fully represent the diversity of imaging protocols, patient populations, and pathological conditions encountered in real clinical practice.

Pathological Coverage: The research primarily addresses general lung segmentation tasks and may not extensively cover specialized pathological conditions that require domain-specific adaptations. Complex cases such as severe pneumothorax, extensive pleural effusions, or post-surgical chest configurations may require additional specialized considerations.

Computational Resources: The development and evaluation of hybrid architectures are constrained by available computational resources, which may limit the exploration of extremely large-scale models or extensive hyperparameter optimization. The research focuses on practically deployable solutions rather than computationally intensive research prototypes.

Clinical Validation: An extensive clinical validation with practicing radiologists and integration into clinical workflows are beyond the current scope. Such validation would require regulatory approval and extensive clinical partnerships.

Real-time Processing: Although computational efficiency is considered, the research does not primarily focus on real-time processing requirements for emergency medical applications. The emphasis is on achieving high accuracy while maintaining reasonable computational efficiency for routine clinical use.

1.5 Thesis Outline

This thesis is organized into several chapters that systematically address the research objectives and provide comprehensive coverage of the investigation:

Chapter 1: Introduction presents the research overview, problem statement, objectives, limitations, and scope, providing the foundational context for the entire investigation.

Chapter 2: Literature Review provides a comprehensive survey of existing approaches in lung segmentation, covering traditional methods, CNN-based approaches, Vision Transformers in medical imaging, hybrid architectures, and current benchmarks. This chapter establishes the theoretical foundation and identifies research gaps that motivate the proposed work.

Chapter 3: Methodology details the proposed hybrid CNN-Transformer architecture, including the overall system design, component specifications, fusion mechanisms, training procedures, and evaluation protocols. This chapter provides sufficient technical detail for reproducibility.

Chapter 4: Results presents comprehensive experimental results, including quantitative performance comparisons, qualitative analysis, ablation studies, and computational efficiency analysis. This chapter demonstrates the effectiveness of the proposed approach.

Chapter 5: Conclusion interprets the research findings, analyzes the implications for clinical practice, discusses limitations and potential improvements, and compares results with existing literature. This chapter provides critical analysis of the research outcomes.

CHAPTER 2
LITERATURE REVIEW

2.1 Traditional Segmentation Methods

The evolution of lung segmentation methods provides valuable context for understanding current hybrid approaches. Early segmentation techniques emerged from computer vision fundamentals, adapting general image processing methods to medical imaging challenges. These traditional approaches, while seemingly outdated, established important principles that continue to influence modern architectures. The four-category framework proposed by Islam and Zhang (2018) includes rule-based models, pixel-based methods, deformable-based models, and registration-based models.

Rule-Based Methods: The earliest systematic attempts at automated lung segmentation relied on rule-based systems that tried to codify anatomical knowledge. These methods operated on the principle that certain image characteristics—such as intensity values, shape properties, and spatial relationships—could be used to identify lung regions. If a region appeared dark, roughly triangular, and located in expected anatomical positions, it likely represented lung tissue. While these systems worked reasonably well for high-quality images with clear boundaries, they struggled significantly with image variability and pathological presentations where expected patterns were disrupted.

Pixel Classification Approaches: These methods treated segmentation as a point-wise labeling problem, where each pixel was independently classified as lung or non-lung tissue based on intensity values and local texture features. Classical machine learning algorithms such as Support Vector Machines (SVMs) and Random Forests were commonly employed for this classification task Çallı et al. (2021). While conceptually straightforward, these methods often produced fragmented results lacking spatial coherence, as they did not consider the spatial relationships between neighboring pixels.

Deformable Models: This category introduced shape constraints into the segmentation process, attempting to fit flexible templates to lung boundaries. Active contour models and level set methods represented popular implementations of this approach Islam and Zhang (2018), Hooper (2023). These methods incorporated prior knowledge about expected lung shapes and could maintain smooth, continuous boundaries. However, they

required careful initialization and often failed when actual lung shapes deviated significantly from templates due to disease or anatomical variations.

Registration-Based Methods: These approaches compared new images against databases of previously segmented examples, warping reference segmentation to match new cases through image registration techniques. Atlas-based segmentation methods exemplified this approach, leveraging multiple reference images to improve robustness Yuan et al. (2024). While this approach could leverage existing expert annotations, it was computationally expensive and struggled with cases that differed substantially from reference datasets.

2.2 CNN-Based Approaches

The introduction of deep learning fundamentally shifted the paradigm from manual feature engineering to automatic feature learning. Convolutional Neural Networks brought several key advantages: they could learn hierarchical representations automatically, handle image variability better than hand-crafted features, and achieve unprecedented accuracy on benchmark datasets.

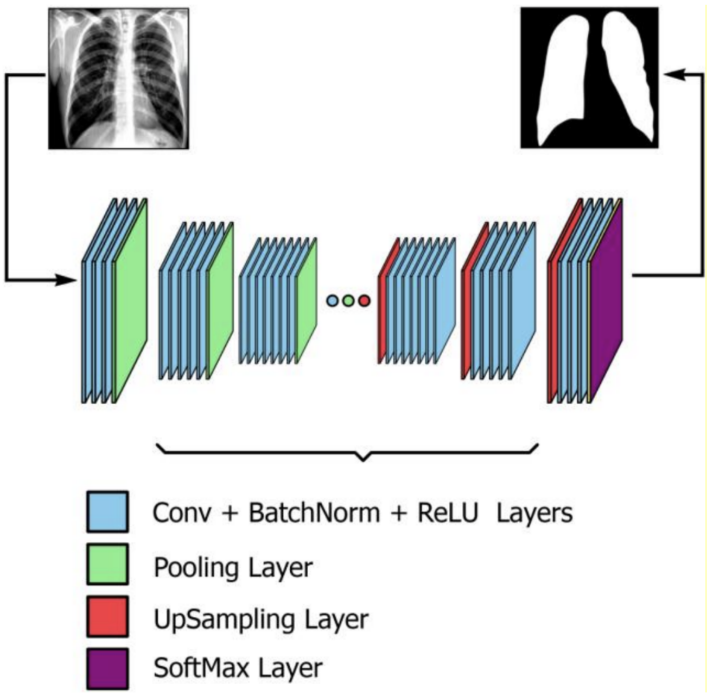


Figure 2.1

Architecture of Deep Encoder-Decoder Convolutional Neural Networks -Kalinovsky and Kovalev (2016)

2.2.1 U-Net and Its Variants

The U-Net architecture, introduced by Ronneberger et al. (2015), became particularly influential in medical image segmentation. Its symmetric encoder-decoder structure with skip connections elegantly addressed a fundamental challenge—how to combine high-level semantic information with low-level spatial details. The encoder pathway progressively reduced spatial resolution while increasing semantic abstraction, learning *what* was present in the image. The decoder pathway reconstructed spatial resolution for precise localization, determining *where* structures were located. Skip connections preserved fine-grained details that might otherwise be lost during the encoding process.

Subsequent innovations built upon U-Net's foundation, addressing specific limitations through architectural modifications. Zhou et al. (2018) introduced U-Net++, adding dense skip connections and nested decoder pathways to improve feature propagation between encoder and decoder layers. This nested architecture created multiple semantic levels, allowing the network to handle objects at different scales more effectively.

Oktay et al. (2018) developed Attention U-Net, incorporating attention gates that allowed the network to focus on relevant regions while suppressing irrelevant features. The attention mechanism helped the model automatically learn to highlight salient features useful for the segmentation task while reducing responses from background regions.

Çiçek et al. (2016) extended the concept to volumetric data with 3D U-Net, adapting the architecture for three-dimensional medical imaging modalities. This extension demonstrated the versatility of the U-Net concept across different dimensionalities and imaging types.

2.2.2 Modern CNN Architectures

Beyond U-Net variants, other CNN architectures have contributed to lung segmentation advancement. DeepLab series introduced atrous convolutions for capturing broader context without losing resolution, addressing the trade-off between receptive field size and computational efficiency Chen et al. (2017). Various backbone networks including ResNet and DenseNet were incorporated to prevent gradient vanishing issues and improve feature learning in deeper networks.

In chest X-ray lung segmentation specifically, CNN-based methods have achieved im-

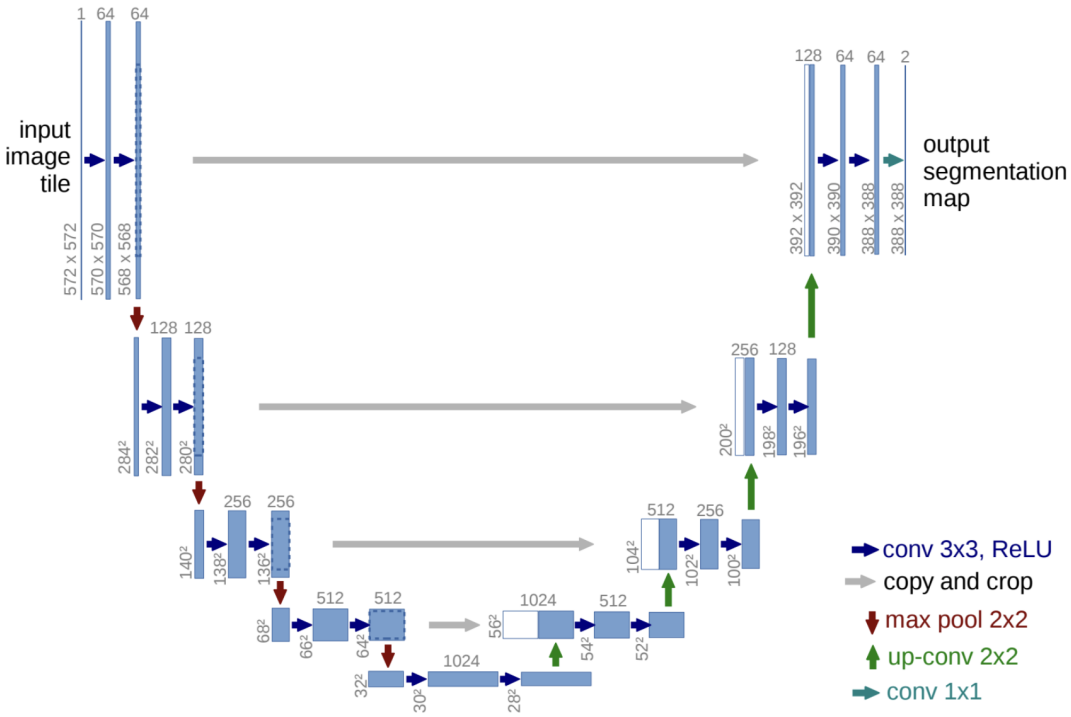


Figure 2.2

Unet architecture overview

pressive results. Recent work has reported Dice scores exceeding 97% on standard datasets. The study by Khalil et al. (2024) achieved 96.2% Dice scores by combining U-Net architectures with the Segment Anything Model, demonstrating the continued effectiveness of CNN-based approaches when enhanced with modern techniques.

Data augmentation techniques became standard practice to address limited training data, with rotation, flipping, zooming, and elastic deformation commonly employed to artificially increase dataset size and improve model generalization.

2.3 Vision Transformers in Medical Imaging

The success of Transformers in natural language processing sparked significant interest in adapting these architectures for computer vision tasks. Vision Transformers process images differently from CNNs—instead of using sliding window convolutions, ViTs divide images into fixed-size patches and model relationships between all patches simultaneously through self-attention mechanisms.

2.3.1 Transformer Fundamentals

The core innovation of Vision Transformers lies in the self-attention mechanism, which allows the model to consider relationships between all image patches simultaneously. Unlike CNNs that build understanding through local receptive fields, Transformers can immediately access global information, making them particularly suitable for tasks requiring long-range dependency modeling.

The standard ViT architecture Dosovitskiy et al. (2020) divides an input image into fixed-size patches, linearly embeds each patch, adds positional encodings, and feeds the resulting sequence to a standard Transformer encoder. This approach treats images as sequences of patches, analogous to how text is treated as sequences of words in natural language processing.

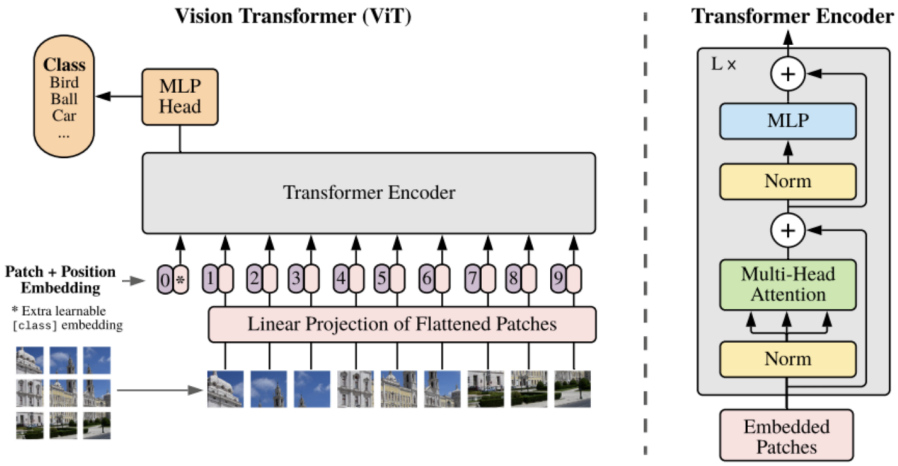


Figure 2.3

Vision Transformer architecture overview

Multi-Head Self-Attention Mechanism

The Multi-Head self-Attention (MHSA) mechanism extends the basic self-attention by running multiple attention functions in parallel. Rather than computing attention once, MHSA processes the input through h different attention heads, each learning different types of relationships.

Given an input sequence $X \in \mathbb{R}^{N \times d}$, each attention head i computes:

$$\text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V) \tag{2.1}$$

where $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d \times d_k}$ are learned projection matrices, and:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2.2}$$

The outputs from all heads are concatenated and projected:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, ..., \text{head}_h)W^O \tag{2.3}$$

This multi-head approach allows the model to attend to different representation subspaces simultaneously, capturing various types of relationships within the input sequence Dosovitskiy et al. (2020).

2.3.2 Medical Imaging Applications

In medical imaging, this global perspective offers unique advantages. As comprehensively reviewed by Shamshad et al. (2023), ViTs excel at capturing long-range dependencies and understanding global image structure. For lung segmentation, this capability is particularly valuable because lung boundaries often depend on understanding the overall chest anatomy, not just local edge patterns.

Pure ViT approaches have shown promising results for chest X-ray analysis. Recent work achieved 97.47% F1 scores for lung segmentation, demonstrating that Transformer architectures can indeed handle segmentation tasks effectively. However, these pure ViT models often require substantial computational resources and may struggle with fine-grained boundary details that are crucial for precise medical segmentation.

Several medical-specific adaptations of Vision Transformers have emerged. Swin Transformer Liu et al. (2021) introduced hierarchical processing with shifted windows, making Transformers more efficient for dense prediction tasks like segmentation. This hierarchical approach mimics the multi-scale processing of CNNs while maintaining Transformer's global modeling capabilities.

2.3.3 Challenges and Limitations

Despite their impressive capabilities, pure ViT models face several challenges in medical imaging applications. They typically require larger datasets for effective training compared to CNNs, which can be problematic in medical domains where annotated data is often limited. The computational requirements are also higher, potentially limiting their applicability in resource-constrained clinical environments.

Moreover, while ViTs excel at global context modeling, they may struggle with fine-grained spatial details that are crucial for precise medical segmentation. The patch-based processing can sometimes miss important boundary information that CNNs naturally capture through their local receptive fields.

2.4 Hybrid CNN-Transformer Architectures

The recognition that CNNs and Transformers have complementary strengths naturally led to hybrid architectures. CNNs excel at local feature extraction and spatial detail preservation, while Transformers provide superior global context modeling and long-range dependency capture. Combining these capabilities represents a logical evolution in architecture design.

2.4.1 Pioneering Hybrid Models

Pioneered this approach with TransUNet, one of the first successful hybrid architectures for medical image segmentation. TransUNet employs a CNN encoder for initial feature extraction, then feeds these features to Transformer layers for global context modeling. The decoder combines both local CNN features and global Transformer representations to produce final segmentation masks. This architecture achieved state-of-the-art results on multiple medical segmentation benchmarks, validating the hybrid concept. Chen et al. (2021)

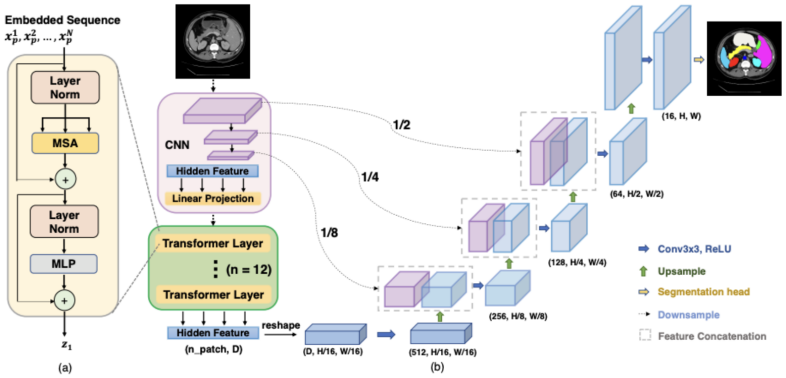


Figure 2.4

TransUnet architecture overview

Zhang et al. (2021) introduced TransFuse, taking a parallel approach where CNN and Transformer branches process the same input simultaneously. Their BiFusion module leverages self-attention mechanisms to selectively combine features from both branches, allowing the network to adaptively choose the most relevant information for each spatial location. This parallel processing approach maintains the independence of both pathways while enabling effective information fusion.

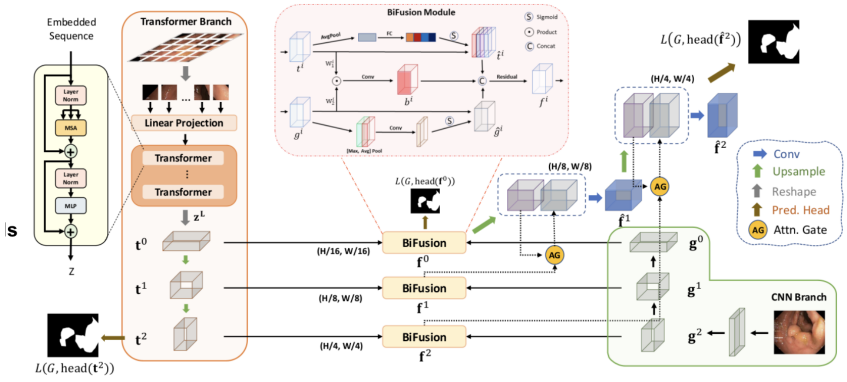


Figure 2.5

TransFuse architecture overview

Gao et al. (2021) proposed UTNet, another hybrid architecture that incorporates Transformer modules within a U-shaped network structure. UTNet demonstrated that Transformer components could be effectively integrated into existing CNN frameworks with-

out completely abandoning proven architectural principles.

2.4.2 Fusion Strategies

Different fusion strategies have been explored for combining CNN and Transformer features. Early concatenation approaches simply combined features from both pathways, but more sophisticated methods have emerged. Attention-based fusion mechanisms allow adaptive weighting of CNN and Transformer features based on their relevance to the current task and spatial location.

Cross-attention mechanisms enable bidirectional information flow between CNN and Transformer pathways, allowing each component to inform and enhance the other. This bidirectional interaction can lead to improved feature representations that leverage the strengths of both architectural paradigms.

Multi-scale fusion approaches combine features at different resolution levels, enabling the hybrid model to benefit from both local and global information across multiple scales. This is particularly important for medical segmentation tasks where relevant information exists at different spatial scales.

2.4.3 Feature Fusion Mechanisms

How CNN and Transformer features are combined might matter as much as the architectures themselves. Zhang et al. (2021) introduced TransFuse with sophisticated BiFusion modules that don't simply concatenate features. Instead, they use self-attention and multi-modal fusion mechanisms to intelligently combine information from both pathways. The BiFusion module incorporates both channel attention (implemented as SE-Block) to promote global information from the Transformer branch and spatial attention (adopted from CBAM) to enhance local details from CNN features while suppressing irrelevant regions.

The fusion process in TransFuse involves several key steps. First, channel attention is applied to Transformer features while spatial attention processes CNN features. Then, a Hadamard product models fine-grained interaction between features from both branches. Finally, the interaction features and attended features are concatenated and passed through residual blocks. This approach achieved superior performance across multiple datasets, with ablation studies showing that BiFusion outperformed simple concatenation plus

residual connections by significant margins.

Yunusa et al. (2024) Academic research has systematically categorized fusion approaches into distinct methodologies, each with specific advantages and optimal use cases:

Early Fusion combines features at the input level, where both CNN and Transformer pathways start with shared representations. This approach enables joint learning from the beginning but might limit each architecture's ability to specialize in their respective strengths. Examples include DETR's integration of CNN feature extraction with Transformer sequence modeling, and LeViT's combination of spatially-efficient convolutional stages early in the process.

Late Fusion keeps the pathways separate until the final decision stage. Each architecture processes information independently, then their outputs are combined for final predictions. While conceptually cleaner and easier to implement, this approach might miss opportunities for architectures to inform each other during intermediate processing stages.

Cross-Attention Fusion represents the most sophisticated approach, using attention mechanisms to dynamically decide how to weight features from each pathway. Some image regions might benefit more from CNN processing, while others require Transformer analysis. Cross-attention learns these preferences automatically, adapting to input characteristics and spatial requirements.

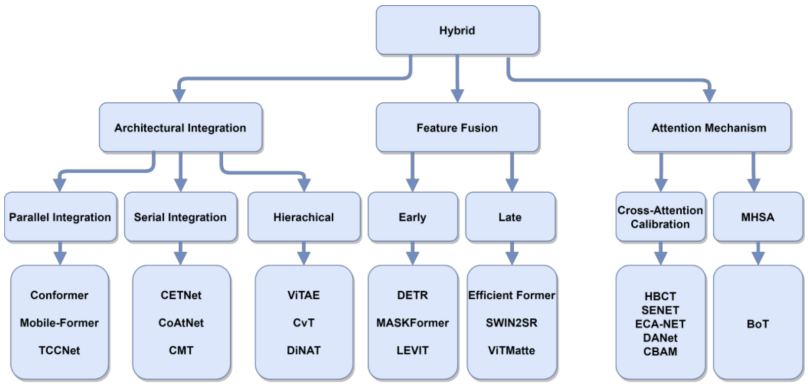


Figure 2.6

Fusion Strategy taxonomy

2.4.4 Attention Mechanism Integration in Hybrid Architectures

Adding attention mechanisms to hybrid architectures significantly enhances their responsiveness and effectiveness. Spatial attention mechanisms help both CNN and Transformer components focus on relevant image regions, creating a feedback loop where each component influences where the other should direct attention.

Fang et al. (2022) developed HBCT for super-resolution tasks, incorporating enhanced spatial attention mechanisms at multiple stages. Their hybrid block integrates spatial attention at both initial and final stages, enhancing spatial features by emphasizing regions of interest while suppressing irrelevant areas. Ablation studies revealed that removing these spatial attention modules significantly impacted performance, with comparative analysis against state-of-the-art models demonstrating HBCT's superior performance while acknowledging the computational complexity trade-offs.

Channel attention adds another crucial dimension to hybrid architectures. The SE-NET approach by Hu et al. (2018) has been successfully adapted for hybrid architectures, helping networks decide which feature channels from CNNs or Transformers are most relevant for specific tasks. This approach involves 'squeeze' (global average pooling) and 'excitation' (dynamic channel recalibration), and has been integrated with various architectures including ResNet-SE, DenseNet-SE, and even ViT-SE variants.

The Convolutional Block Attention Module (CBAM) provides another sophisticated integration approach, stacking channel attention and spatial attention in series. CBAM sequentially infers channel attention vectors and spatial attention maps, decoupling these mechanisms for computational efficiency while leveraging spatial global information through global pooling operations. This modular approach allows seamless integration into various CNN architectures with negligible additional computational cost.

Recent developments in attention-based fusion include Pre-gating and Contextual Attention Gate (PCAG) methods that filter non-informative interactions, and frequency-aware fusion techniques that integrate frequency components in hybrid architectures, representing the cutting edge of fusion methodology research.

2.4.5 Application Gaps in Chest X-ray Segmentation

Despite the success of hybrid CNN-Transformer models in various computer vision tasks, their application to chest X-ray lung segmentation reveals interesting patterns. Most hybrid innovations in chest imaging have focused on classification tasks rather than segmentation applications. Studies have developed sophisticated hybrid models for COVID-19 detection, lung disease classification, and other diagnostic tasks, but segmentation applications remain relatively limited.

This gap is particularly noteworthy given that lung segmentation requires both fine-grained boundary detection (CNN strength) and global anatomical context understanding (Transformer strength). The projection-based nature of chest X-rays, with overlapping anatomical structures and variable pathological presentations, suggests that hybrid models might be particularly well-suited for this application.

2.5 Recent Advances and State-of-the-Art Methods

2.5.1 SAM-based Approaches

The Segment Anything Model (SAM) has emerged as a significant development in medical image segmentation. Khalil et al. (2024) developed an innovative approach using YOLO to generate prompts for SAM, creating an automatic lung segmentation pipeline that achieved impressive results with Dice scores of 96.2% through cross-dataset validation strategies.

The proposed SAM-YOLO framework demonstrates the potential of leveraging foundation models for medical imaging tasks. The approach works by first using YOLO v5 to detect lung fields and generate bounding boxes, which then serve as prompts for SAM's segmentation capabilities. The integration involves several preprocessing techniques including contrast-limited adaptive histogram equalization (CLAHE) and gamma correction to enhance lung boundaries and improve discrimination between lungs and surrounding tissues.

What makes this approach particularly interesting is how it leverages foundation models that weren't specifically trained for medical images. SAM's zero-shot generalization capabilities, when properly prompted through YOLO's detection results, achieve performance comparable to specialized medical segmentation models. The framework addresses computational efficiency concerns by using YOLO's efficient single-stage de-

tection followed by SAM's focused segmentation, resulting in reduced computational burden compared to processing entire images.

The study's comprehensive evaluation across multiple datasets (JSRT, Montgomery County, Shenzhen, and others) demonstrates remarkable robustness and generalization capability against variability in sensors, populations, disease manifestations, and imaging conditions. However, the authors acknowledge that the framework's effectiveness heavily depends on YOLO's accuracy in lung region detection, as inaccurate bounding boxes can degrade overall performance.

Recent developments have also explored combining SAM with other detection models and different prompting strategies, suggesting that this foundation model approach represents a promising direction for medical image segmentation that warrants further investigation in hybrid architecture contexts.

2.6 Current Benchmarks and Evaluation Methods

2.6.1 Standard Datasets

Standard evaluation of lung segmentation methods typically employs well-established datasets that provide consistent benchmarks for comparing different approaches. The Montgomery County dataset includes 138 images from tuberculosis screening programs, providing diversity in pathological presentations. The Shenzhen dataset offers additional variation with 662 images from Chinese medical institutions. And Qatar-COVID19 chest X-ray database for COVID-19 detection.

These datasets provide ground truth segmentation created by expert radiologists, enabling quantitative comparison of different methods. However, each dataset has its own characteristics in terms of image quality, population demographics, and pathological representation, which affects the generalization of results across different clinical contexts.

2.6.2 Evaluation Metrics

Overlap-Based Metrics

Dice Coefficient remains the most widely used metric in medical image segmentation Hooper (2023); Jadon (2020). This overlap-based metric appears to be particularly

robust for class imbalance scenarios common in medical imaging. The mathematical formulation is:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} \tag{2.4}$$

where Y represents the ground truth mask, $\hat{Y}$ denotes the predicted segmentation mask, $|Y \cap \hat{Y}|$ is the intersection (overlap) between ground truth and predicted masks, and $|Y|$, $|\hat{Y}|$ represent the cardinalities of the respective sets Yuan et al. (2024). The coefficient ranges from 0 to 1, with higher values indicating better segmentation performance Turk and Kılıçaslan (2025).

Intersection over Union (IoU) IoU, also known as the Jaccard Index, provides a complementary perspective to the Dice coefficient. Mathematical relationship between Dice and IoU may suggest that both metrics capture similar aspects of segmentation quality:

$$\text{IoU}(Y, \hat{Y}) = \frac{|Y \cap \hat{Y}|}{|Y \cup \hat{Y}|} \tag{2.5}$$

where Y is the ground truth mask, $\hat{Y}$ is the predicted segmentation mask, $|Y \cap \hat{Y}|$ represents the intersection of both masks, and $|Y \cup \hat{Y}|$ represents their union. The relationship between these metrics follows: $\text{IoU} = \frac{\text{Dice}}{2-\text{Dice}}$ Rahman et al. (2020). IoU appears to be more sensitive to segmentation quality in small objects compared to Dice coefficient.

Distance-Based Metrics

Hausdorff Distance Hausdorff Distance (HD) measures the maximum distance between predicted and ground truth boundaries, providing insight into boundary accuracy that overlap-based metrics might miss Hooper (2023). The formulation is:

$$\text{HD}(A, B) = \max\{h(A, B), h(B, A)\} \tag{2.6}$$

where the directed Hausdorff distance is defined as:

$$h(A, B) = \max_{a \in A} \min_{b \in B} ||a - b|| \tag{2.7}$$

where A and B represent the boundary point sets of ground truth and predicted segmentation respectively, a and b are individual boundary points, and $||a - b||$ denotes the Euclidean distance between points.

HD95 (95th Percentile Hausdorff Distance) HD95 appears to be more robust than standard Hausdorff distance as it considers the 95th percentile rather than the absolute maximum, reducing sensitivity to outliers Yao et al. (2024):

$$\text{HD95} = \max_{k95\%}(h(A, B), h(B, A)) \tag{2.8}$$

where $h(A, B) = \max_{a \in A} \min_{b \in B} ||a - b||$, $h(B, A) = \max_{b \in B} \min_{a \in A} ||a - b||$, A and B are boundary point sets, and $k95\%$ indicates the 95th percentile of the distance distribution

Performance-Based Metrics

Accuracy Accuracy represents the overall proportion of correctly classified pixels in the segmentation task:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2.9}$$

where TP denotes true positives (correctly identified Region of Interest pixels), TN represents true negatives (correctly identified non-ROI pixels), FP indicates false positives (non-ROI pixels incorrectly classified as ROI), and FN refers to false negatives (ROI pixels incorrectly classified as non-ROI).

Sensitivity and Specificity Sensitivity (true positive rate) measures the proportion of actual lung pixels correctly identified by the segmentation method:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{2.10}$$

Specificity (true negative rate) measures the proportion of non-lung pixels correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{2.11}$$

where TP, TN, FP, and FN have the same definitions as in the accuracy formula. These metrics provide complementary perspectives on segmentation performance, particularly important in clinical applications Yuan et al. (2024).

2.6.3 Loss Functions

Distribution-Based Loss Functions

Cross-Entropy Loss Binary cross-entropy serves as the foundation for many segmentation loss functions. The pixel-wise formulation is:

$$H(Y_{i,j}, \hat{P}_{i,j}) = -[Y_{i,j} \log(\hat{P}_{i,j}) + (1 - Y_{i,j}) \log(1 - \hat{P}_{i,j})] \tag{2.12}$$

where $Y_{i,j}$ represents the ground truth label at pixel position (i, j) (0 or 1), $\hat{P}_{i,j}$ denotes the predicted probability at pixel position (i, j), and $\log$ is the natural logarithm. However, cross-entropy loss is likely to be sensitive to class imbalances, which are common in segmentation tasks Hooper (2023).

Focal Loss Focal Loss addresses class imbalance by down-weighting easy examples and focusing on hard negatives Lin et al. (2017):

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{2.13}$$

where p_t is the model's estimated probability for the ground truth class, $\gamma > 0$ controls the focusing parameter (typically $\gamma = 2$), and α_t is a weighting factor that balances positive/negative examples Jadon (2020).

Region-Based Loss Functions

Dice Loss Dice Loss directly optimizes the Dice coefficient during training:

$$DL(y, \hat{p}) = 1 - \frac{2y\hat{p} + 1}{y + \hat{p} + 1} \tag{2.14}$$

where y represents the ground truth binary mask, $\hat{p}$ denotes the predicted probability map, and the addition of 1 in numerator and denominator ensures numerical stability when $y = \hat{p} = 0$ Jadon (2020).

Distance-Based Loss Functions

Hausdorff Distance Loss Distance-based loss functions appear to be more sensitive to detailed boundary characteristics but can be unstable during training Hooper (2023):

$$HDL(Y_{i,j}, \hat{P}_{i,j}) = (Y_{i,j} - \hat{P}_{i,j})^2 (d^{\eta}_{Y_{i,j}} + d^{\eta}_{\hat{P}_{i,j}}) \tag{2.15}$$

where $Y_{i,j}$ is the ground truth label at pixel (i, j), $\hat{P}_{i,j}$ is the predicted probability at pixel (i, j), $d_{Y_{i,j}}$ and $d_{\hat{P}_{i,j}}$ represent minimum distances to ground truth and predicted boundaries respectively, and η is a hyperparameter that controls the emphasis on boundary regions Karimi and Salcudean (2019).

Compound Loss Functions

Combo Loss Combo Loss combines Dice Loss with modified cross-entropy to leverage benefits of both approaches:

$$CL(y, \hat{y}) = \alpha L_{m-bce} - (1 - \alpha) DL(y, \hat{y}) \tag{2.16}$$

where L_{m-bce} represents modified binary cross-entropy Taghanaki et al. (2019). The loss functions

Combination tailored-made Loss There are many loss functions have been used right off-the-shelf in segmentation tasks, also many tailored-made ones for a specific image data type, case study and objective. Ma et al. (2021)

2.6.4 Performance Evaluation Considerations

Evaluation of lung segmentation methods involves several considerations that may influence metric selection. Distance-based metrics typically appear more suitable for applications where detailed boundary characteristics are important, while overlap-based metrics provide general alignment assessment Hooper (2023). The choice of evaluation metrics should align with the specific clinical application requirements.

Research suggests that combining multiple loss functions from different categories (distribution-based, overlap-based, and distance-based) often encourages stronger segmentation performance Jadon (2020). This appears particularly relevant for lung segmentation where both global shape and precise boundary delineation are clinically significant.

Contemporary approaches likely benefit from careful consideration of class imbalance, with focal-based loss functions showing promise for highly imbalanced datasets typical in medical imaging Lin et al. (2017). The stability of distance-based loss functions can be improved by combining them with overlap-based metrics during training optimization.

CHAPTER 3
METHODOLOGY

3.1 Overview

This chapter presents the methodological framework for developing a hybrid CNN-Vision Transformer architecture specifically designed for chest X-ray lung segmentation. The approach appears to address the critical gap identified in the literature review, where existing methods either excel in local feature extraction (CNNs) or global context modeling (Vision Transformers) but struggle to effectively combine both capabilities for segmentation tasks.

The methodology encompasses five key components: dataset preparation and annotation, the proposed hybrid architecture design, loss function optimization, evaluation metrics, and experimental validation strategies. This systematic approach may suggest a comprehensive solution to the limitations observed in current segmentation approaches.

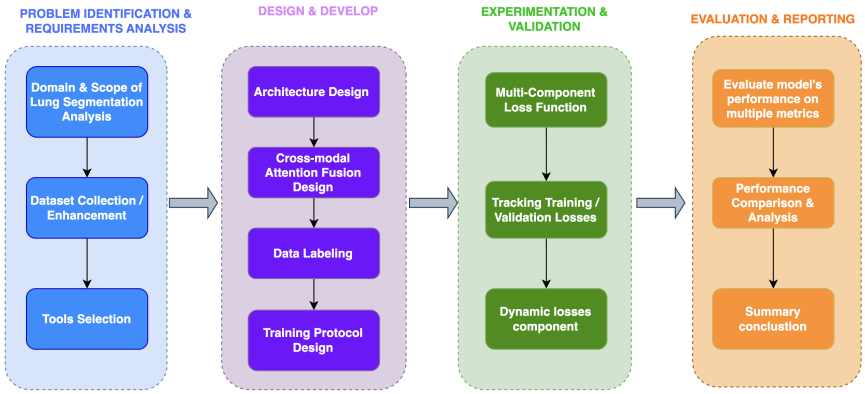


Figure 3.1

Methodology Diagram

3.2 Dataset Introduction

3.2.1 Data Sources and Composition

The experimental dataset comprises a carefully curated collection of 800 chest X-ray images from two well-established public medical imaging repositories: the Montgomery County dataset and the Shenzhen Hospital dataset Jaeger et al. (2014). This combination appears strategically chosen to provide diverse imaging conditions, pathological variations, and demographic characteristics essential for training a generalizable lung segmentation model.

Montgomery County Dataset: This dataset originates from the tuberculosis control program administered by the Department of Health and Human Services of Montgomery County, Maryland, United States. The collection includes 138 posterior-anterior chest radiographs with a balanced distribution of 80 normal cases and 58 images exhibiting various degrees of tuberculosis manifestations, providing valuable pathological diversity for model training.

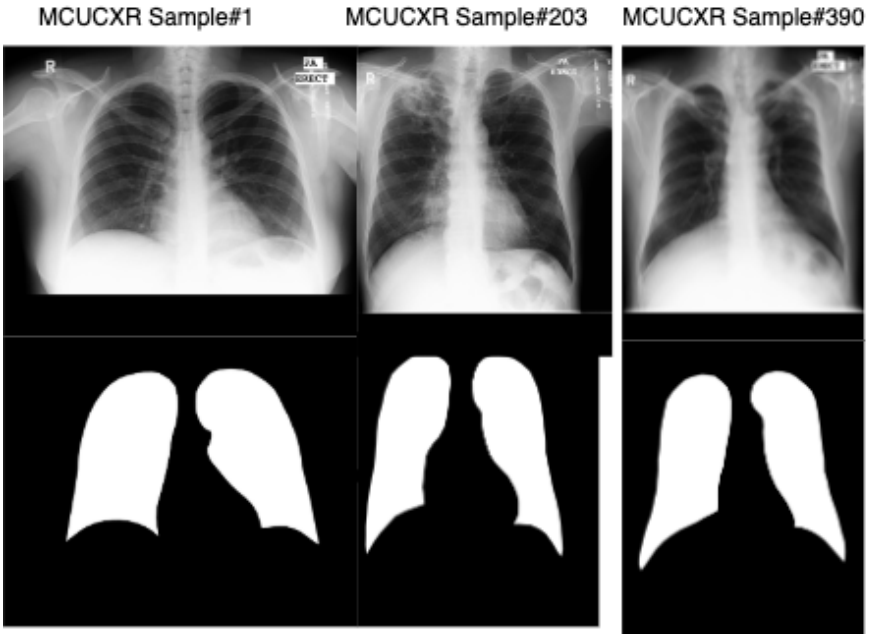


Figure 3.2

MCU samples images & ground truth masks

Shenzhen Hospital Dataset: Collected from the Shenzhen No.3 People's Hospital, Guangdong Medical College in China, this dataset contributes 662 frontal chest X-rays obtained from outpatient department patients. The distribution includes 326 normal cases and 336 tuberculosis-positive images. This international source appears crucial for enhancing cross-population generalizability and imaging protocol diversity.

Qatar-COVID19 Dataset: Rahman et al. (2021) Researchers from Qatar University, University of Dhaka, and collaborators in Pakistan and Malaysia have developed a chest X-ray database for COVID-19 detection. The dataset includes images of COVID-19 cases, normal lungs, and other lung infections. It contains 3,616 COVID-19 cases, 10,192 normal images, 6,012 lung opacity cases, and 1,345 viral pneumonia images with lung masks. The database continues to expand as new images become available.

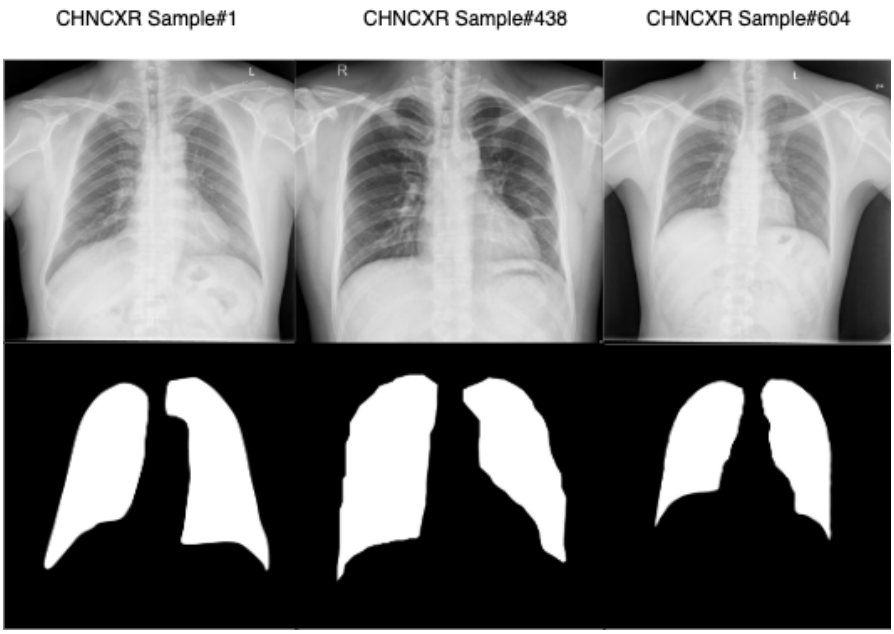


Figure 3.3

SHZ samples images & ground truth masks

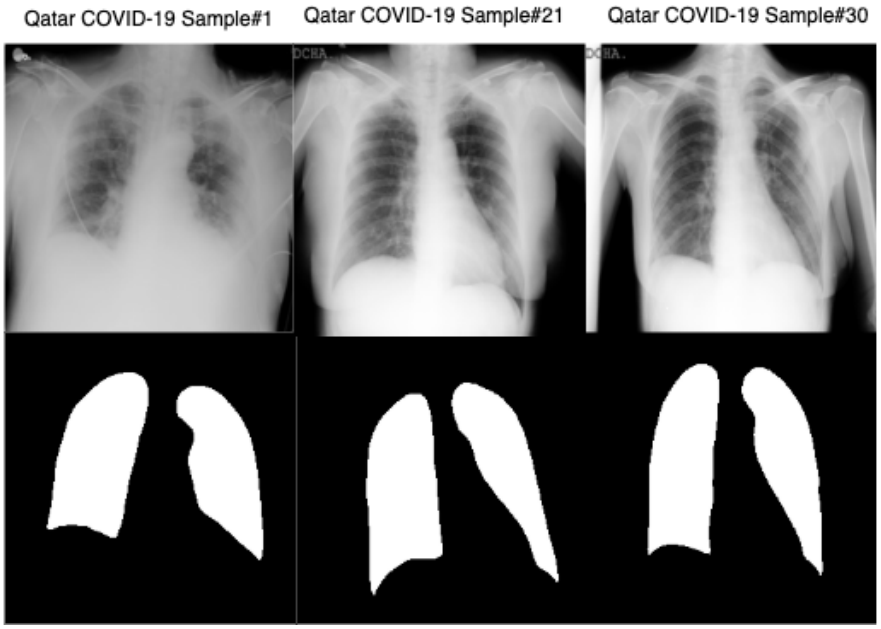


Figure 3.4

Qatar-COVID19 samples images & ground truth masks

3.2.2 Technical Image Specifications

Resolution and Format Characteristics: The datasets exhibit significant variation in image resolution that reflects real-world clinical diversity. Montgomery County images are captured in 12-bit grayscale format with high resolution dimensions of 4020 × 4892 pixels, stored in DICOM format which preserves medical imaging metadata. In contrast, Shenzhen images show variable dimensions with average sizes around 3000 × 3000 pixels, provided in PNG format for broader accessibility. In addition, the Qatar-COVID 19 dataset has 299 x 299 pixels provided in PNG format.

Image Quality Variations: The substantial resolution differences between datasets (ranging from approximately 3000×3000 to 4020×4892 pixels) appear to simulate the heterogeneous imaging conditions encountered in clinical practice. This variation includes different X-ray equipment specifications, exposure parameters, and image acquisition protocols that may affect segmentation performance.

Pathological Diversity: Both datasets encompass chest radiographs from patients with varying degrees of lung abnormalities, including normal anatomical presentations and tuberculosis manifestations of different severities. This pathological diversity is crucial for training segmentation algorithms that maintain accuracy across different disease states and anatomical variations.

3.2.3 Dataset Partitioning and Validation Strategy

Training-Validation-Test Split: The dataset is systematically divided following a strategic distribution designed to maximize both training effectiveness and evaluation reliability. The primary split allocates 80% for training, 10% for validation, and 10% for testing, ensuring sufficient data for model optimization while preserving independent evaluation sets.

Balanced Distribution Maintenance: The partitioning strategy ensures that both normal and pathological cases are proportionally represented across training, validation, and test sets. This balanced approach is likely to prevent bias toward specific pathological conditions and enhance model generalizability across different clinical presentations.

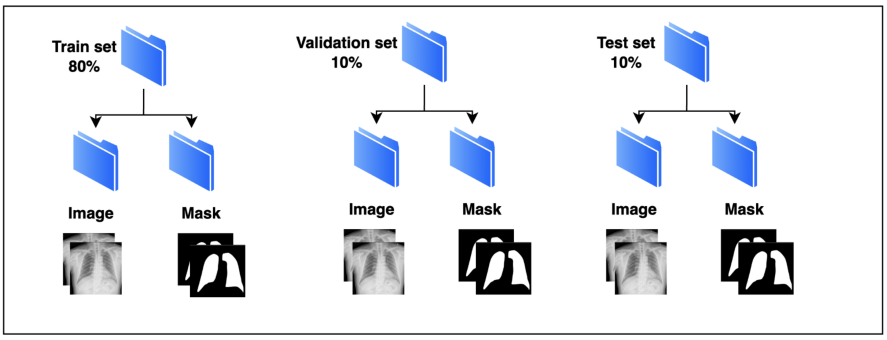


Figure 3.5

Images and Masks folder setup

3.2.4 Dataset Quality Assurance and Limitations

Ground Truth Reliability: The availability of expert-annotated ground truth masks for 704 out of 800 images provides a solid foundation for supervised learning. The remaining 96 images from the Shenzhen dataset initially lacked segmentation masks, necessitating additional annotation efforts using professional annotation tools.

3.3 Labeling Method

3.3.1 Ground Truth Annotation Process

The labeling methodology follows a rigorous manual annotation protocol developed in collaboration with the original research teams and medical imaging experts from the University of Ukraine. This multi-institutional approach appears to ensure annotation quality and clinical relevance of the ground truth masks.

Among the 800 collected images, 704 possess corresponding expert-annotated lung masks that serve as ground truth references. The remaining 96 images from the Shenzhen dataset initially lacked segmentation masks, necessitating additional annotation efforts. The labeling process utilizes Label Studio, a professional annotation platform that provides precise pixel-level marking capabilities essential for segmentation tasks.

3.3.2 Annotation Output

Binary masks delineate lung regions with pixel values of 1 representing lung areas and 0 indicating background areas. The masks preserve anatomical boundaries while excluding cardiac structures.

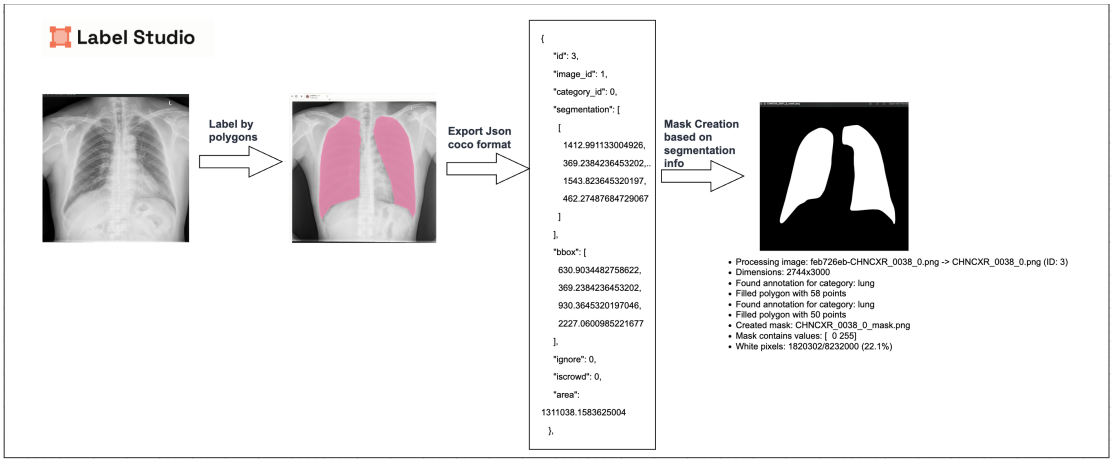


Figure 3.6

Labeling Process in Label Studio

3.4 Proposed Hybrid Model Architecture

3.4.1 Architectural Design

The proposed hybrid model addresses the fundamental limitation where pure CNN approaches miss global anatomical context while Vision Transformers struggle with fine boundary details. The architecture employs a parallel processing strategy that simultaneously extracts local features through convolutional operations and captures global dependencies via transformer attention mechanisms.

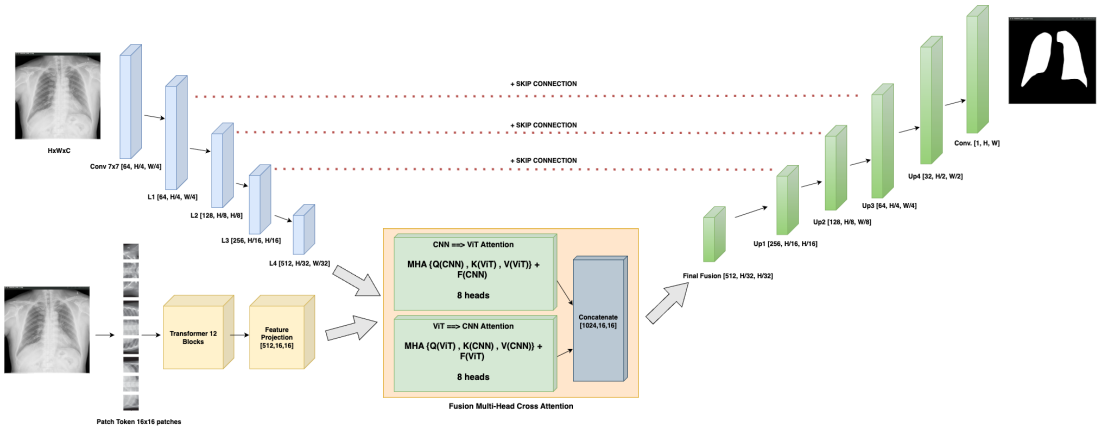


Figure 3.7

Proposed Hybrid Model Architecture

Let $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ represent the input chest X-ray image where H, W, and C denote height, width, and channels respectively. The hybrid architecture can be formulated as:

$$\mathbf{Y} = \mathcal{F}_{fusion}(\mathcal{F}_{CNN}(\mathbf{X}), \mathcal{F}_{ViT}(\mathbf{X})) \tag{3.1}$$

where $\mathcal{F}_{CNN}$ and $\mathcal{F}_{ViT}$ represent the CNN and Vision Transformer pathways respectively, and $\mathcal{F}_{fusion}$ denotes the cross-modal fusion mechanism.

Unlike sequential approaches such as TransUNet that process features through CNN followed by Transformer stages, this parallel design appears to preserve the complementary strengths of both architectures throughout the feature extraction process. The model consists of two primary pathways operating simultaneously on the input data.

3.4.2 CNN Pathway Design

The convolutional pathway utilizes a lightweight CNN encoder based on ResNet architecture with BasicBlock residual connections. The CNN encoder progressively extracts hierarchical features through four main layers with channel progression: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.

The initial feature extraction begins with a 7×7 convolutional layer:

$$\mathbf{F}_0 = \text{MaxPool}(\text{ReLU}(\text{BN}(\text{Conv}_{77}(\mathbf{X})))) \tag{3.2}$$

Each BasicBlock in the residual layers computes:

$$\mathbf{F}_{i,j}^{CNN} = \mathcal{H}(\mathbf{F}_{i,j-1}^{CNN}) + \mathbf{F}_{i,j-1}^{CNN} \tag{3.3}$$

where $\mathcal{H}$ represents the residual function:

$$\mathcal{H}(\mathbf{x}) = \text{BN}(\text{Conv}_{33}(\text{ReLU}(\text{BN}(\text{Conv}_{33}(\mathbf{x}))))) \tag{3.4}$$

The complete CNN pathway produces feature maps at multiple scales:

$$\mathbf{F}_1^{CNN} = \text{Layer1}(\mathbf{F}_0) \in \mathbb{R}^{64 \times H/4 \times W/4} \tag{3.5}$$

$$\mathbf{F}_2^{CNN} = \text{Layer2}(\mathbf{F}_1^{CNN}) \in \mathbb{R}^{128 \times H/8 \times W/8} \tag{3.6}$$

$$\mathbf{F}_3^{CNN} = \text{Layer3}(\mathbf{F}_2^{CNN}) \in \mathbb{R}^{256 \times H/16 \times W/16} \tag{3.7}$$

$$\mathbf{F}_4^{CNN} = \text{Layer4}(\mathbf{F}_3^{CNN}) \in \mathbb{R}^{512 \times H/32 \times W/32} \tag{3.8}$$

3.4.3 Vision Transformer Pathway

The Vision Transformer component employs a lightweight architecture with 12 transformer blocks and embedding dimension $D = 192$. The input image is divided into patches of size $P = 16$, creating $N = \frac{HW}{P^2} = \frac{256 \times 256}{16^2} = 256$ patches.

The patch embedding layer converts image patches to feature vectors:

$$\mathbf{x}_p^i = \text{Conv}_{1616,s=16}(\mathbf{X}_{patch}^i) \in \mathbb{R}^{192} \tag{3.9}$$

The transformer processes the sequence through 12 identical blocks, each containing:

$$\mathbf{z}_l^{'} = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \tag{3.10}$$

$$\mathbf{z}_l = \text{MLP}(\text{LN}(\mathbf{z}_l^{'})) + \mathbf{z}_l^{'} \tag{3.11}$$

where MSA is Multi-Head Self-Attention with the standard formulation:

$$\text{MSA}(\mathbf{z}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O \tag{3.12}$$

Each attention head computes:

$$\text{head}_i = \text{Attention}(\mathbf{z}\mathbf{W}_i^Q, \mathbf{z}\mathbf{W}_i^K, \mathbf{z}\mathbf{W}_i^V) \tag{3.13}$$

The MLP block follows the standard transformer design:

$$\text{MLP}(\mathbf{x}) = \text{Linear}(\text{GELU}(\text{Linear}(\mathbf{x}))) \tag{3.14}$$

with hidden dimension expansion factor of 4 ($192 \rightarrow 768 \rightarrow 192$).

The final ViT features are projected to match CNN dimensions:

$$\mathbf{F}^{ViT} = \text{Conv}_{11}(\text{Reshape}(\mathbf{z}_{12})) \in \mathbb{R}^{512 \times 16 \times 16} \tag{3.15}$$

3.4.4 Cross-Attention Fusion Strategy

The fusion mechanism implements bidirectional cross-attention between CNN and ViT features. The CrossAttentionFusion module computes:

$$\mathbf{F}_{cnn \to vit} = \text{MultiHeadAttn}(\mathbf{F}^{CNN}, \mathbf{F}^{ViT}, \mathbf{F}^{ViT}) + \mathbf{F}^{CNN} \tag{3.16}$$

$$\mathbf{F}_{vit \to cnn} = \text{MultiHeadAttn}(\mathbf{F}^{ViT}, \mathbf{F}^{CNN}, \mathbf{F}^{CNN}) + \mathbf{F}^{ViT} \tag{3.17}$$

Layer normalization is applied to both cross-attended features:

$$\tilde{\mathbf{F}}_{cnn} = \text{LayerNorm}(\mathbf{F}_{cnn \to vit}) \tag{3.18}$$

$$\tilde{\mathbf{F}}_{vit} = \text{LayerNorm}(\mathbf{F}_{vit \to cnn}) \tag{3.19}$$

The final fusion combines both pathways through concatenation and convolution:

$$\mathbf{F}_{fused} = \text{ReLU}(\text{BN}(\text{Conv}_{33}(\text{Concat}(\tilde{\mathbf{F}}_{cnn}, \tilde{\mathbf{F}}_{vit})))) \tag{3.20}$$

This reduces the concatenated features from 1024 to 512 channels:

$$\mathbf{F}_{fused} \in \mathbb{R}^{512 \times H/32 \times W/32} \tag{3.21}$$

3.5 Decoder Architecture and Output Generation

3.5.1 SimpleDecoder Design

The decoder module implements a progressive upsampling strategy to reconstruct the segmentation mask from the fused features. The decoder consists of four upsampling blocks with skip connections from the CNN encoder:

$$\mathbf{D}_1 = \text{UpBlock}_1(\mathbf{F}_{fused}, \mathbf{F}_3^{CNN}) \in \mathbb{R}^{256\times H/16\times W/16} \tag{3.22}$$

$$\mathbf{D}_2 = \text{UpBlock}_2(\mathbf{D}_1, \mathbf{F}_2^{CNN}) \in \mathbb{R}^{128\times H/8\times W/8} \tag{3.23}$$

$$\mathbf{D}_3 = \text{UpBlock}_3(\mathbf{D}_2, \mathbf{F}_1^{CNN}) \in \mathbb{R}^{64\times H/4\times W/4} \tag{3.24}$$

$$\mathbf{D}_4 = \text{UpBlock}_4(\mathbf{D}_3, \mathbf{F}_0) \in \mathbb{R}^{32\times H/2\times W/2} \tag{3.25}$$

Each upsampling block follows the pattern:

$$\text{UpBlock}(\mathbf{x}, \mathbf{skip}) = \text{ReLU}(\text{BN}(\text{Conv}_{33}(\text{ReLU}(\text{BN}(\text{ConvTranspose}_{22,s=2}(\text{Concat}(\mathbf{x}, \mathbf{skip}))))))) \tag{3.26}$$

The final output generation involves two stages:

$$\mathbf{F}_{final} = \text{ConvTranspose}_{22,s=2}(\mathbf{D}_4) \in \mathbb{R}^{16\times H\times W} \tag{3.27}$$

$$\mathbf{Y} = \sigma(\text{Conv}_{11}(\mathbf{F}_{final})) \in \mathbb{R}^{1\times H\times W} \tag{3.28}$$

where σ represents the sigmoid activation function producing probability maps for lung segmentation.

3.5.2 Complete Architecture Flow

The complete hybrid architecture can be summarized as:

$$\mathbf{Y} = \mathcal{D}(\mathcal{F}_{cross}(\mathcal{F}_{CNN}(\mathbf{X}), \mathcal{F}_{ViT}(\mathbf{X}))) \tag{3.29}$$

where:

- $\mathcal{F}_{CNN}$: LightCNNEncoder producing multi-scale features
- $\mathcal{F}_{ViT}$: LightViTEncoder with 12 transformer blocks
- $\mathcal{F}_{cross}$: CrossAttentionFusion with bidirectional attention
- $\mathcal{D}$: SimpleDecoder with progressive upsampling

3.5.3 Addressing Current Limitations

The proposed architecture directly targets several key limitations identified in existing segmentation approaches. Traditional CNN-based methods, while effective for boundary detection, often lack the global context necessary for maintaining anatomical consistency across large lung regions. Conversely, pure Vision Transformer approaches may struggle with precise boundary localization despite their superior global modeling capabilities.

Current hybrid models predominantly focus on disease classification tasks rather than segmentation-specific applications. This research gap appears significant, as segmentation requires both fine-grained boundary accuracy and global anatomical understanding—requirements that classification tasks do not necessarily demand.

3.5.4 Advantages of the Proposed Approach

The parallel processing design offers several potential advantages over existing sequential hybrid approaches. By processing features simultaneously through both pathways, the model may preserve information that could be lost in sequential processing stages. The cross-modal attention mechanism enables dynamic adaptation to different image regions and pathological conditions.

The chest X-ray specific optimization represents another key advantage. Unlike general-purpose medical imaging models, this architecture is tailored to the unique characteristics of chest radiography, including typical lung morphology, common pathological patterns, and imaging artifacts specific to this modality.

3.5.5 Potential Limitations and Considerations

Despite its theoretical advantages, the proposed architecture may face certain limitations. The parallel processing approach likely increases computational complexity compared to simpler CNN-only models. Training stability might be challenging due to the need to optimize both CNN and Transformer components simultaneously.

The fusion mechanism, while potentially powerful, adds architectural complexity that could lead to overfitting, particularly with limited training data. Careful regularization and validation strategies appear necessary to ensure the model generalizes effectively to unseen data.

3.6 Loss Function Design

3.6.1 Multi-Component Loss Strategy

Medical image segmentation presents unique challenges that no single loss function can adequately address. The proposed approach employs a combined loss function that addresses multiple aspects of segmentation quality: class imbalance, boundary accuracy, overlap optimization, and training stability.

The total loss function is formulated as a weighted combination of four complementary components:

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{BCE} + \beta\mathcal{L}_{Dice} + \gamma\mathcal{L}_{Focal} + \delta\mathcal{L}_{Boundary} \tag{3.30}$$

where α, β, γ, and δ are dynamic weighting parameters that adapt throughout training phases.

3.6.2 Component Loss Functions

Binary Cross-Entropy Loss serves as the foundation component, providing stable gradients and proven convergence properties:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \tag{3.31}$$

where y_i and $\hat{y}_i$ represent ground truth and predicted probabilities respectively.

Dice Loss directly optimizes the Dice coefficient, enabling direct optimization of the evaluation metric:

$$\mathcal{L}_{Dice} = 1 - \frac{2|\mathbf{Y} \cap \hat{\mathbf{Y}}|}{|\mathbf{Y}| + |\hat{\mathbf{Y}}|} = 1 - \frac{2\sum_{i=1}^{N} y_i\hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon} \tag{3.32}$$

where ϵ is a smoothing parameter to prevent division by zero.

Focal Loss addresses severe class imbalance by down-weighting easy examples and focusing on hard cases:

$$\mathcal{L}_{Focal} = -\frac{1}{N}\sum_{i=1}^{N}\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{3.33}$$

where $p_t = \hat{y}_i$ if $y_i = 1$, otherwise $p_t = 1 - \hat{y}_i$, α_t balances positive/negative examples, and γ is the focusing parameter (typically $\gamma = 2$).

Boundary Loss emphasizes accurate edge delineation through distance transform weighting:

$$\mathcal{L}_{Boundary} = \frac{1}{N}\sum_{i=1}^{N}(\phi_G(x_i)\cdot\hat{y}_i + \phi_{\hat{G}}(x_i)\cdot y_i) \tag{3.34}$$

where ϕ_G and $\phi_{\hat{G}}$ represent distance transforms of ground truth and predicted boundaries respectively.

3.6.3 Dynamic Weighting Strategy

The combination weights follow an adaptive training strategy that evolves across training phases:

$$\begin{bmatrix}\alpha\\ \beta\\ \gamma\\ \delta\end{bmatrix} = \begin{cases}[0.6, 0.2, 0.2, 0.0] & \text{if epoch} \leq 0.4 \times \text{total_epochs}\\ [0.3, 0.3, 0.2, 0.2] & \text{if } 0.4 < \text{epoch} \leq 0.8 \times \text{total_epochs}\\ [0.2, 0.3, 0.1, 0.4] & \text{if epoch} > 0.8 \times \text{total_epochs}\end{cases} \tag{3.35}$$

This adaptive weighting strategy optimizes learning progression by emphasizing stability in early training, balanced learning in mid-training, and boundary refinement in late training phases. In addition, also planning for unweighted 4 multi-losses configuration during training for evaluation and comparison the various protocol.

3.7 Evaluation Metrics

3.7.1 Overlap-Based Metrics

Dice Coefficient serves as the primary evaluation metric, measuring the overlap between predicted and ground truth segmentations:

$$\text{Dice}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{2|\mathbf{Y} \cap \hat{\mathbf{Y}}|}{|\mathbf{Y}| + |\hat{\mathbf{Y}}|} = \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{3.36}$$

where $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ represent ground truth and predicted binary masks respectively.

Intersection over Union (IoU) provides complementary overlap assessment:

$$\text{IoU}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{|\mathbf{Y} \cap \hat{\mathbf{Y}}|}{|\mathbf{Y} \cup \hat{\mathbf{Y}}|} = \frac{\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i - \sum_{i=1}^{N} y_i \hat{y}_i} \tag{3.37}$$

The relationship between Dice and IoU can be expressed as: $\text{Dice} = \frac{2 \times \text{IoU}}{1 + \text{IoU}}$.

3.7.2 Boundary-Specific Metrics

Hausdorff Distance quantifies the maximum distance between predicted and true boundaries:

$$HD(\mathbf{Y}, \hat{\mathbf{Y}}) = \max \left\{ \max_{y \in \partial \mathbf{Y}} \min_{\hat{y} \in \partial \hat{\mathbf{Y}}} d(y, \hat{y}), \max_{\hat{y} \in \partial \hat{\mathbf{Y}}} \min_{y \in \partial \mathbf{Y}} d(\hat{y}, y) \right\} \tag{3.38}$$

where $\partial \mathbf{Y}$ and $\partial \hat{\mathbf{Y}}$ denote boundary pixels, and $d(\cdot, \cdot)$ represents Euclidean distance.

3.7.3 Clinical Relevance Metrics

Sensitivity (Recall) measures the proportion of actual lung pixels correctly identified:

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i} \tag{3.39}$$

Specificity quantifies correct background pixel identification:

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{\sum_{i=1}^{N} (1 - y_i)(1 - \hat{y}_i)}{\sum_{i=1}^{N} (1 - y_i)} \tag{3.40}$$

Precision evaluates segmentation accuracy:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} \hat{y}_i} \tag{3.41}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives respectively.

3.8 Experimental Design and Validation

3.8.1 Training Protocol

The training methodology employs a systematic approach with careful hyperparameter selection and validation strategies. The dataset division follows a 80:10:10 ratio for training, validation, and testing, ensuring sufficient data for model optimization while preserving independent evaluation sets. The training process conducts on using Google Colab with A100 GPU.

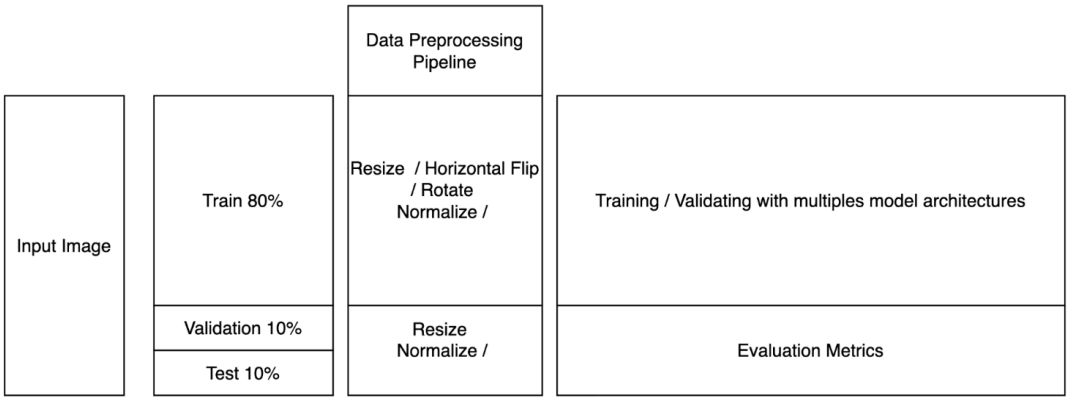


Figure 3.8

Training Method

3.8.2 Ablation Studies

Comprehensive ablation studies are planned to evaluate individual component contributions to overall performance. These studies include: loss function component analysis (BCE, Dice, Focal, Boundary), Vision Transformer patch size optimization, CNN pathway isolation, and Transformer pathway isolation.

The ablation approach is likely to provide valuable insights into which architectural components contribute most significantly to segmentation performance and where future optimization efforts should focus.

3.8.3 Baseline Comparisons

The evaluation framework includes comparisons with established segmentation approaches including traditional U-Net, Vision Transformer variants, and existing hybrid methods. This comprehensive comparison appears essential for demonstrating the practical value

of the proposed approach.

Performance comparisons will utilize identical training data, evaluation metrics, and experimental conditions to ensure fair assessment of architectural improvements rather than implementation differences.

CHAPTER 4
EXPERIMENT

4.1 Experiment Configurations

This chapter presents the outcomes from our hybrid CNN-Vision Transformer architecture for chest X-ray lung segmentation. We conducted three distinct training scenarios to evaluate how model performance changes under different data conditions and architectural configurations. Each scenario was designed with specific objectives: establishing baseline performance, assessing generalization capability, and examining the impact of training data diversity.

4.1.1 Training Scenarios

Scenario 01: Controlled Baseline Evaluation

The first scenario used only the combined MC and SHZ datasets (640 training images, 80 validation images, 80 test images). Training and testing were performed on the same data distribution. The main purpose here was to establish baseline performance under controlled conditions, where the model encounters similar patterns during both training and testing phases. This scenario helps us understand the upper performance limit when domain shift is minimal.

Scenario 02: Cross-Dataset Generalization

The second scenario maintained the same training and validation split from MC and SHZ datasets (640 training, 80 validation). However, testing was conducted on a completely unseen dataset—the Qatar-COVID-19 dataset containing 3,616 chest X-ray images from a different imaging protocol and patient population. This cross-dataset evaluation was crucial for assessing how well our model generalizes to different imaging conditions and protocols, which is essential for real-world clinical application.

Scenario 03: Enhanced Training with Diverse Data

The third scenario expanded the training data by incorporating normal lung images, resulting in 1,437 training images and 280 validation images. Testing remained on the Qatar-COVID-19 dataset (3,616 images). The purpose was to assess whether increasing

training data diversity, particularly with more varied normal lung presentations, could improve the model's ability to generalize to new datasets and handle a wider range of anatomical variations.

Each scenario was evaluated using multiple architectural configurations to identify the optimal configuration.

4.1.2 Architectural Variants

For each training scenario, we tested several architectural variants:

- **Hybrid model with Loss (BCE + Dice):** The complete architecture combining CNN and ViT pathways with cross-attention fusion, trained using a combination of Binary Cross-Entropy and Dice loss.
- **Hybrid model with 4 adaptive losses:** The hybrid architecture incorporating four adaptive loss functions that dynamically weight different segmentation objectives during training.
- **Hybrid model with 4 unweighted losses:** The hybrid architecture incorporating four unweighted loss functions.
- **CNN pathway only with Loss (BCE + Dice):** Only the convolutional pathway active, without cross-fusion from the ViT branch.
- **ViT pathway only with Loss (BCE + Dice):** Only the Vision Transformer pathway active, without utilizing CNN features.

4.2 Experiment Results

4.2.1 Scenario 01: Baseline Performance on MC+SHZ Dataset

The first scenario establishes baseline performance when training and testing occur on similar data distributions. Table 4.1 presents the quantitative results across different architectural configurations on the MC+SHZ test set.

Accuracy Analysis (98.32%–98.38%)

All four configurations achieved tightly clustered accuracy between 98.32% and 98.38% (0.06 percentage point range), indicating pixel-level accuracy provides limited discrimination for lung segmentation. This occurs because most pixels represent correctly classified background regions. The Hybrid Adaptive model achieved the highest accuracy

(98.38 $\pm$ 0.16%) with notably lower standard deviation than BCE+Dice ($\pm$ 0.16% vs $\pm$ 0.31%), suggesting more stable performance.

Counterintuitively, CNN-only (98.33%) performed nearly identically to baseline Hybrid (98.32%), while ViT-only (98.35%) exceeded both. This underscores why accuracy alone inadequately captures boundary delineation quality or handling of challenging anatomical regions.

Dice Score Analysis (96.49%–96.65%)

Dice score showed more meaningful differentiation, though the 0.16 percentage point range remained narrow. Hybrid Adaptive achieved the highest score (96.65 $\pm$ 0.54%), outperforming BCE+Dice by 0.16 points. Critically, standard deviation decreased from $\pm$ 0.75% to $\pm$ 0.54%, indicating adaptive loss weighting stabilizes performance across varying difficulty levels.

ViT-only achieved 96.60% Dice, remarkably close to the best hybrid and exceeding CNN-only (96.52%). For in-domain evaluation with well-matched distributions, the ViT's global lung shape consistency nearly compensates for weaker local boundary sensitivity. The global attention mechanism appears to maintain anatomically plausible contours when local features are ambiguous.

CNN-only's lower Dice (96.52%) reflects a trade-off: reduced sensitivity (95.70%) for higher specificity (99.17%), adopting a conservative strategy that under-segments lung regions to avoid false positives.

IoU Analysis (93.22%–93.52%)

The more stringent IoU metric showed slightly larger differences. Hybrid Adaptive achieved 93.52 $\pm$ 1.01%, a 0.30 point improvement over baseline (93.22 $\pm$ 1.40%), with reduced standard deviation demonstrating adaptive loss stabilization.

IoU's sensitivity to boundary precision stems from its severe penalization of both false positives and negatives. All models exceeding 93% indicates excellent overlap, though

the 3–3.5 point gap between Dice and IoU scores (e.g., 96.65% vs 93.52%) reflects IoU's squared penalty amplifying segmentation errors.

CNN-only (93.28%) falls between hybrid models: strong local boundary detection but occasional small errors accumulate without global context. ViT-only (93.43%) benefits from global shape consistency but lacks CNN's fine-grained boundary refinement.

Sensitivity Analysis (95.70%–96.50%)

Sensitivity revealed more substantial architectural differences. ViT-only achieved highest sensitivity (96.50 $\pm$ 0.37%), logically consistent with the transformer's global attention mechanism capturing all lung tissue regions despite slightly imperfect boundary precision.

CNN-only exhibited lowest sensitivity (95.70 $\pm$ 0.42%), 0.43 points below best hybrid (96.13%) and 0.8 points below ViT-only. This conservative behavior likely stems from local feature detection focus: when evidence is ambiguous (heavy rib shadowing, cardiac border), the CNN misclassifies actual lung pixels as non-lung to maintain high specificity.

Both hybrid models achieved balanced sensitivity (96.11–96.13%), with cross-attention fusion successfully combining ViT's higher recall with CNN's boundary precision. Adaptive loss maintained comparable sensitivity to BCE+Dice (96.13% vs 96.11%) while reducing variance ($\pm$ 0.35% vs $\pm$ 0.71%), indicating consistent performance across anatomical variations.

Specificity Analysis (98.95%–99.17%)

All models achieved remarkably high specificity above 98.95%, expected given background regions' distinct intensity characteristics. However, subtle differences reveal architectural trade-offs.

CNN-only achieved highest specificity (99.17 $\pm$ 0.16%), rarely including non-lung regions, logically pairing with its lower sensitivity—a conservative strategy potentially

advantageous when false positives trigger unnecessary clinical procedures. ViT-only showed lowest specificity (98.95 $\pm$ 0.21%), consistent with highest sensitivity, occasionally including peripheral regions beyond true boundaries. Higher standard deviation suggests less stable boundary localization across varying anatomical contexts.

Hybrid Adaptive achieved 99.11% specificity with excellent stability ($\pm$ 0.16%), effectively balancing CNN's high specificity and ViT's high sensitivity through cross-attention fusion considering both local features and global context.

Hausdorff Distance Analysis (19.17–19.68 pixels)

HD measures maximum boundary distance, capturing "worst-case" error. Values around 19–20 pixels indicate the model's boundary can deviate approximately 2 cm from true lung edge in challenging regions.

Hybrid Adaptive achieved lowest HD (19.17 $\pm$ 2.29), improving over baseline Hybrid (19.68 $\pm$ 3.38). More importantly, standard deviation decreased by approximately 1 pixel, suggesting adaptive loss weighting prevents occasional large boundary errors.

CNN-only showed HD of 19.53 $\pm$ 2.82, while ViT-only achieved 19.40 $\pm$ 2.04. ViT's slightly better HD suggests global shape consistency helps avoid catastrophic boundary errors despite imperfect local precision. CNN alone occasionally misidentified boundaries in complex regions (costophrenic angles, cardiac border), causing larger maximum errors despite good average performance.

HD95 Analysis (8.23–9.65 pixels)

HD95 provides more stable boundary error measurement by excluding worst 5% of outlier distances, better reflecting typical accuracy. Values of 8.23–9.65 pixels indicate 95% of boundary points fall within approximately 8–10 pixels of true boundary.

Hybrid Adaptive achieved best HD95 (8.23 $\pm$ 1.23 pixels), a 0.47 pixel improvement over baseline Hybrid (8.70 $\pm$ 1.83 pixels). CNN-only exhibited worst HD95 (9.65 $\pm$ 2.55 pixels) with notably higher standard deviation, suggesting the CNN pathway with-

out global context guidance struggles with consistent boundary localization despite good average metrics.

ViT-only achieved HD95 of 8.55 ± 1.70 pixels, substantially better than CNN-only and competitive with baseline Hybrid. This reinforces that transformer-learned global shape priors maintain reasonable boundary estimates when local features are ambiguous, preventing large systematic boundary errors from CNN-only approaches.

Summary of Model Comparison

Synthesizing across metrics, Hybrid Adaptive demonstrates modest but consistent improvements over simpler configurations. Gains are most apparent in prediction stability (lower standard deviations) and boundary accuracy (HD, HD95) rather than overlap metrics (Dice, IoU), suggesting hybrid fusion and adaptive loss weighting primarily handle challenging cases more gracefully rather than dramatically improving average-case performance.

The surprisingly strong ViT-only performance (second-best Dice and HD95) challenges CNN superiority assumptions for segmentation. On in-domain data with matched distributions, transformer global reasoning appears nearly sufficient, though boundary precision benefits from CNN local feature extraction. CNN-only's conservative, high-specificity approach suggests value in applications prioritizing false positive minimization over sensitivity.

Table 4.1

Performance comparison for models trained and tested on the MC+SHZ dataset. Best results are highlighted in bold.

Model Configuration	**Acc (%)**	**Dice (%)**	**IoU (%)**	**Sens (%)**	**Spec (%)**	**HD**	**HD95**
Hybrid (BCE+Dice Losses)	98.32 ± 0.31	96.49 ± 0.75	93.22 ± 1.40	96.11 ± 0.71	99.02 ± 0.23	19.68 ± 3.38	8.70 ± 1.83
Hybrid (Adaptive Losses)	$\mathbf{98.38 \pm 0.16}$	$\mathbf{96.65 \pm 0.54}$	$\mathbf{93.52 \pm 1.01}$	96.13 ± 0.35	$\mathbf{99.11 \pm 0.16}$	$\mathbf{19.17 \pm 2.29}$	$\mathbf{8.23 \pm 1.23}$
CNN only (Adaptive Losses)	98.33 ± 0.17	96.52 ± 0.57	93.28 ± 1.05	95.70 ± 0.42	$\mathbf{99.17 \pm 0.16}$	19.53 ± 2.82	9.65 ± 2.55
ViT only (Adaptive Losses)	98.35 ± 0.21	96.60 ± 0.65	93.43 ± 1.21	$\mathbf{96.50 \pm 0.37}$	98.95 ± 0.21	19.40 ± 2.04	8.55 ± 1.70

4.2.2 Scenario 02: Cross-Dataset Generalization to Qatar-COVID-19

The second scenario evaluates the model's ability to generalize to a completely different dataset with distinct imaging protocols and patient demographics. Table 4.2 presents performance on the Qatar-COVID-19 test set when trained on MC+SHZ data.

Accuracy Analysis (96.32%–97.38%)

Accuracy ranged from 96.32% to 97.38%, representing only a 1 percentage point drop from Scenario 01. Hybrid BCE+Dice achieved highest accuracy (97.38 $\pm$ 1.13%), but this modest decrease masks significant challenges evident in other metrics.

Standard deviation increased dramatically across all models. CNN-only showed $\pm$ 1.66% variation compared to $\pm$ 0.17% in Scenario 01, suggesting inconsistent performance across different Qatar-COVID-19 images. Some images might have characteristics similar to training data yielding good performance, while others present sufficient novelty causing considerable degradation. The ranking shifted notably: ViT-only (97.17%) now substantially outperforms CNN-only (96.32%), whereas they were nearly equivalent in Scenario 01, hinting that ViT's global reasoning may generalize better across domain shifts than CNN's local feature detection.

Dice Score Analysis (92.27%–94.56%)

Dice scores revealed substantial performance degradation. Hybrid BCE+Dice achieved 94.56 $\pm$ 2.15%, approximately a 2 percentage point drop from Scenario 01 (96.65%). Standard deviation increased from $\pm$ 0.54% to $\pm$ 2.15%, indicating considerable performance variation across images, likely reflecting sensitivity to novel imaging characteristics or pathological presentations.

Critically, ViT-only (94.14 $\pm$ 2.35%) now performs competitively with hybrid models, nearly matching Hybrid Adaptive (94.34 $\pm$ 1.97%). The gap narrowed from approximately 0.5 points in Scenario 01 to just 0.2 points. Meanwhile, CNN-only dropped to 92.27 $\pm$ 3.37%, falling 2 points behind ViT-only. This reversal suggests CNNs' local feature detectors, while excellent for in-domain precision, may be more fragile when

confronting distributional shifts.

Counterintuitively, Hybrid Adaptive (94.34%) performed slightly worse than BCE+Dice (94.56%), contrary to expectations. The adaptive loss strategy that excelled at stabilizing in-domain performance may not have adapted appropriately to Qatar dataset characteristics, highlighting that loss functions optimized for one distribution may not transfer optimally.

IoU Analysis (85.82%–89.75%)

IoU's stringent evaluation revealed pronounced degradation. Best model (Hybrid BCE+Dice at 89.75 $\pm$ 3.81%) showed a 3.77 point drop from Scenario 01. The Dice-IoU gap widened from approximately 3 points to nearly 5 points, indicating segmentation errors became more consequential under domain shift.

CNN-only IoU plummeted to 85.82 $\pm$ 5.70%, with standard deviation substantially higher than Scenario 01 ($\pm$ 1.05%). This massive variability suggests the CNN performs reasonably on some images but fails substantially on others with novel appearance characteristics. ViT-only achieved 89.02 $\pm$ 4.13%, now substantially outperforming CNN-only by over 3 points—a dramatic shift from Scenario 01 where these architectures performed comparably. The transformer's attention mechanism appears more robust to changes in low-level visual statistics.

The three hybrid configurations cluster closely (88.91%–89.75%), suggesting fusion strategy variations matter less than the fundamental architectural choice to combine pathways. Standard deviations remain elevated but lower than CNN-only, indicating hybrid fusion provides stability benefits.

Sensitivity Analysis (89.67%–92.52%)

Sensitivity dropped substantially, ranging from 89.67% to 92.52% compared to 95.70%–96.50% in Scenario 01. This 4–7 point degradation indicates models frequently fail to identify actual lung pixels, leading to under-segmentation.

CNN-only suffered most severe loss, dropping to 89.67 $\pm$ 6.54%. The substantially elevated standard deviation (from $\pm$ 0.42% in Scenario 01) reveals highly inconsistent performance, likely reflecting CNN's reliance on learned texture and edge patterns that vary significantly between datasets. ViT-only maintained relatively better sensitivity (92.17 $\pm$ 4.45%), though still substantially below Scenario 01 (96.50%). The transformer's global attention helps maintain awareness of lung regions even when local features differ.

All three hybrid models achieved remarkably similar sensitivity (92.08%–92.52%), suggesting fusion successfully leverages ViT's global context to maintain reasonable recall. Hybrid BCE+Dice achieved highest sensitivity (92.52 $\pm$ 4.19%), though elevated variance indicates inconsistent performance.

Specificity Analysis (98.60%–99.03%)

Specificity remained relatively high (98.60%–99.03%), experiencing much smaller degradation than sensitivity. This asymmetry makes clinical sense: background regions have fairly consistent characteristics regardless of imaging protocol, while lung tissue appearance varies substantially with acquisition parameters.

Hybrid models maintained specificity (99.02–99.03%) nearly identical to Scenario 01, with BCE+Dice showing perfect consistency (99.02% in both scenarios). This stability contrasts sharply with meaningful sensitivity drops, reflecting that avoiding false positives remains achievable under domain shift, while capturing all lung tissue becomes challenging when visual characteristics change.

CNN-only specificity (98.60 $\pm$ 0.35%) represents the lowest value, paired with worst sensitivity (89.67%), suggesting the CNN struggles fundamentally with the new domain rather than adopting a conservative strategy. Standard deviations remained remarkably low ($\pm$ 0.20% to $\pm$ 0.37%), indicating background classification stability.

Precision Analysis (95.34%–96.80%)

Precision ranged from 95.34% to 96.80%, indicating when models predict a pixel as lung tissue, they're usually correct. Both hybrid BCE+Dice and Adaptive achieved highest precision (96.80%), showing fusion mechanisms help make confident predictions.

The precision-recall trade-off becomes evident: CNN-only achieved 95.34% precision but only 89.67% sensitivity, suggesting conservative behavior—accurate when identifying lung but missing substantial regions. ViT-only balanced slightly better precision (96.33%) with higher sensitivity (92.17%).

Low standard deviations ($\pm$ 0.80% to $\pm$ 1.18%) indicate consistent behavior when making positive predictions. This stability, combined with much higher variance in sensitivity, suggests the primary challenge under domain shift is missing actual lung tissue (false negatives) rather than incorrectly including non-lung regions (false positives).

Hausdorff Distance Analysis (33.17–50.84 pixels)

HD revealed perhaps the most alarming degradation. Values ranging from 33.17 to 50.84 pixels represent worst-case boundary errors substantially exceeding the 19–20 pixels in Scenario 01—approximately doubling, indicating that on some images, segmentation boundaries diverge drastically from ground truth.

CNN-only exhibited catastrophically poor HD (50.84 $\pm$ 16.39 pixels) with enormous variance. The $\pm$ 16.39 pixel standard deviation means worst errors might be 34 pixels in some images while exceeding 67 pixels in others, suggesting complete failure on certain Qatar-COVID-19 images with substantially different contrast characteristics.

ViT-only performed considerably better (40.60 $\pm$ 13.76 pixels), though still representing poor absolute performance. The transformer's global reasoning helps avoid complete boundary localization failure, though significant errors occur. Surprisingly, Hybrid Unweighted achieved best HD (33.17 $\pm$ 9.88 pixels), outperforming BCE+Dice (34.05 pixels). However, Hybrid Adaptive showed worse HD (43.18 $\pm$ 10.38 pixels), suggesting adaptive loss functions optimized on training data characteristics may amplify certain errors when those characteristics shift.

HD95 Analysis (16.87–26.69 pixels)

HD95 increased substantially from 8–10 pixels in Scenario 01 to 17–27 pixels, indicating typical boundary errors approximately doubled under domain shift. Even excluding worst 5% of outliers, most boundary predictions now deviate by 17–27 pixels from ground truth—clinically significant displacement affecting diagnostic interpretation.

CNN-only performed worst (26.69 $\pm$ 12.14 pixels) with massive variance indicating wildly inconsistent boundary localization. ViT-only achieved 19.74 $\pm$ 9.79 pixels, substantially better than CNN but still representing significant degradation from Scenario 01 (8.55 pixels). Hybrid models clustered between 16.87–18.43 pixels, with BCE+Dice achieving best HD95 (16.87 $\pm$ 8.09 pixels).

Examining variance stability, Hybrid Adaptive achieved lowest standard deviation ($\pm$ 7.00 pixels), making it the most consistent performer despite not having absolute best mean score. This stability suggests certain hybrid configurations generalize more predictably across domains, valuable in clinical deployment where consistent performance may be preferable to occasionally better but unpredictable results.

Summary of Domain Shift Effects

Cross-dataset evaluation reveals several critical patterns. First, performance degradation is highly metric-dependent: accuracy dropped minimally (1 point), Dice moderately (2 points), IoU substantially (3–4 points), and boundary metrics catastrophically (HD doubled). This hierarchical degradation reflects that coarse-level accuracy is preserved while fine-grained boundary precision suffers dramatically.

Second, architectural robustness rankings shifted significantly. ViT-only, which underperformed CNN-only on in-domain data, became substantially superior under domain shift, suggesting transformers' learned spatial relationships generalize better than CNNs' local feature detectors when visual statistics change. However, hybrid fusion remained superior to either pathway alone, indicating complementary robustness benefits.

Third, variance increased dramatically across all metrics compared to Scenario 01. This elevated inconsistency poses major clinical deployment challenges—even if average per-

formance seems acceptable, high variance means substantial failures occur on a meaningful fraction of images.

Finally, Hybrid BCE+Dice generally outperformed Adaptive configuration contrary to Scenario 01 patterns, suggesting loss functions optimized for one domain may not transfer optimally. This highlights the importance of developing domain-agnostic training strategies rather than domain-specific optimizations that may harm generalization.

Table 4.2

Cross-dataset generalization performance. Models were trained on MC+SHZ and tested on the unseen Qatar-COVID-19 dataset.

Model	**Acc (%)**	**Dice (%)**	**IoU (%)**	**Sens (%)**	**Spec (%)**	**Prec (%)**	**HD**	**HD95**
CNN-Only	96.32 ± 1.66	92.27 ± 3.37	85.82 ± 5.70	89.67 ± 6.54	98.60 ± 0.35	95.34 ± 1.18	50.84 ± 16.39	26.69 ± 12.14
ViT-Only	97.17 ± 1.25	94.14 ± 2.35	89.02 ± 4.13	92.17 ± 4.45	98.87 ± 0.37	96.33 ± 1.12	40.60 ± 13.76	19.74 ± 9.79
Hybrid (BCE+Dice Losses)	$\mathbf{97.38 \pm 1.13}$	$\mathbf{94.56 \pm 2.15}$	$\mathbf{89.75 \pm 3.81}$	$\mathbf{92.52 \pm 4.19}$	99.02 ± 0.27	$\mathbf{96.80 \pm 0.87}$	$\mathbf{34.05 \pm 11.44}$	$\mathbf{16.87 \pm 8.09}$
Hybrid (Adaptive Losses)	97.27 ± 1.06	94.34 ± 1.97	89.35 ± 3.51	92.08 ± 3.90	$\mathbf{99.03 \pm 0.20}$	$\mathbf{96.80 \pm 0.80}$	43.18 ± 10.38	18.43 ± 7.00
Hybrid (Unweighted Losses)	97.14 ± 1.11	94.09 ± 2.08	88.91 ± 3.67	92.46 ± 4.28	98.74 ± 0.24	95.91 ± 0.96	33.17 ± 9.88	17.04 ± 7.24

4.2.3 Scenario 03: Enhanced Training with Diverse Data

The third scenario investigates whether expanding training data diversity improves generalization. Models were trained on MC+SHZ combined with additional normal lung images (1,437 training images, 280 validation images) and tested on Qatar-COVID-19. Table 4.3 presents the results.

Accuracy Analysis (98.01%–98.38%)

Accuracy recovered impressively, ranging from 98.01% to 98.38%—remarkably close to Scenario 01 (98.32–98.38%) and substantially improved from Scenario 02 (96.32–97.38%). Hybrid Unweighted achieved highest accuracy (98.38 $\pm$ 0.73%), matching Scenario 01's best result. This near-complete recovery demonstrates training data diversity can largely overcome domain shift challenges at pixel classification level.

Standard deviations decreased dramatically compared to Scenario 02, clustering around $\pm$ 0.73% to $\pm$ 1.02% versus $\pm$ 1.06% to $\pm$ 1.66% previously. This variance reduction is perhaps more significant than mean improvements—enhanced training data not only improves average performance but also stabilizes predictions across diverse Qatar-COVID-19 images.

Interestingly, CNN-only (98.08 $\pm$ 1.02%) now performs nearly identically to ViT-only (98.01 $\pm$ 0.86%), reversing Scenario 02's gap where ViT substantially outperformed CNN. This convergence suggests that with sufficient training diversity, both architectural approaches learn adequately robust representations.

Dice Score Analysis (95.96%–96.71%)

Dice scores improved dramatically from Scenario 02's range of 92.27–94.56% to 95.96–96.71%. Hybrid Unweighted achieved best Dice (96.71 $\pm$ 1.32%), nearly recovering to Scenario 01's best performance (96.65%). This represents approximately 2 percentage points improvement over Scenario 02, closing roughly two-thirds of the domain shift performance gap.

Standard deviations decreased substantially, with most models showing $\pm$ 1.32% to

± 1.92% variation compared to ± 1.97% to ± 3.37% in Scenario 02—approximately halving variance, indicating much more consistent segmentation quality across different Qatar-COVID-19 images.

Fascinatingly, CNN-only (96.06 ± 1.92%) now slightly outperforms ViT-only (95.96 ± 1.56%), reversing their Scenario 02 relationship where ViT achieved 94.14% versus CNN's 92.27%. This 2-point swing suggests CNNs benefit more dramatically from expanded training diversity. The convolutional pathway, when exposed to sufficient examples of local feature variations, learns more robust texture and edge detectors that generalize effectively.

The tight clustering of all three hybrid configurations (96.51–96.71% Dice) indicates that with abundant diverse training data, specific loss function strategy matters minimally. The 0.2 percentage point spread suggests researchers should prioritize data collection over loss function engineering when resources allow.

IoU Analysis (92.28%–93.67%)

IoU values recovered from Scenario 02's range of 85.82–89.75% to 92.28–93.67%, representing 3–4 percentage point improvements. Hybrid Unweighted achieved 93.67 ± 2.43% IoU, actually exceeding Scenario 01's best result (93.52%) by 0.15 points. This surpassing of in-domain performance demonstrates that training data diversity doesn't merely overcome domain shift—it can produce models that generalize even better than those optimized for specific distributions.

Variance reduction is striking: standard deviations decreased from ± 3.51% to ± 5.70% in Scenario 02 to ± 2.43% to ± 3.49% here. CNN-only's variance dropped most dramatically, from ± 5.70% to ± 3.49%, suggesting CNNs' brittleness to domain shift can be substantially mitigated through diverse training examples.

Both single-pathway models (CNN: 92.48%, ViT: 92.28%) achieved nearly identical IoU scores, remarkably improved from Scenario 02 (85.82% and 89.02% respectively). CNN gained 6.66 points while ViT gained 3.26 points—double the improvement, explaining why CNN-only now matches ViT-only performance.

Sensitivity Analysis (94.66%–96.42%)

Sensitivity improved substantially from Scenario 02's 89.67–92.52% range to 94.66–96.42%, representing 3–5 percentage point gains. Hybrid Unweighted achieved highest sensitivity (96.42 $\pm$ 2.35%), approaching Scenario 01's best result (96.50%) and recovering most recall lost to domain shift.

Variance reductions are equally impressive. CNN-only's sensitivity variance dropped from $\pm$ 6.54% to $\pm$ 3.73%, and ViT-only improved from $\pm$ 4.45% to $\pm$ 2.63%. These decreases indicate expanded training diversity helps models consistently identify lung regions across varied Qatar-COVID-19 presentations.

CNN-only's sensitivity (94.66 $\pm$ 3.73%) improved by approximately 5 points from Scenario 02 (89.67%)—the largest gain among all architectures. This dramatic recovery supports the hypothesis that CNNs' poor cross-domain recall in Scenario 02 stemmed from limited exposure to texture and contrast variations during training.

Specificity Analysis (98.85%–99.36%)

Specificity remained extremely high across all models (98.85–99.36%), comparable to both previous scenarios. This consistency confirms background classification remains largely invariant to training data quantity—the fundamental task of distinguishing lung regions from background doesn't benefit much from additional examples.

Hybrid BCE+Dice achieved highest specificity (99.36 $\pm$ 0.16%), marginally exceeding even Scenario 01's best result (99.11%). This slight improvement, paired with strong sensitivity (95.21%), indicates excellent balance between recall and precision.

CNN-only achieved very high specificity (99.25 $\pm$ 0.20%) but showed lowest sensitivity (94.66%), reflecting the CNN pathway's conservative strategy. Conversely, Hybrid Unweighted balanced slightly lower specificity (99.06%) with highest sensitivity (96.42%), indicating a more liberal approach that captures lung regions comprehensively.

Precision Analysis (96.39%–97.92%)

Precision ranged from 96.39% to 97.92%, substantially improved from Scenario 02 (95.34–96.80%). Hybrid BCE+Dice achieved highest precision (97.92 $\pm$ 0.56%), indicating when the model predicts lung tissue, it's correct over 97% of the time.

The uniformly low standard deviations ($\pm$ 0.56% to $\pm$ 1.36%) demonstrate consistent positive prediction accuracy across diverse Qatar-COVID-19 images. ViT-only showed lowest precision (96.39 $\pm$ 1.36%) paired with strong sensitivity (95.58%), consistent with its tendency toward more liberal segmentation.

Hausdorff Distance Analysis (20.28–27.32 pixels)

HD values improved dramatically from Scenario 02's catastrophic range of 33.17–50.84 pixels to 20.28–27.32 pixels. Hybrid Unweighted achieved best HD (20.28 $\pm$ 9.29 pixels), representing approximately 13 pixels (nearly 40%) improvement over its Scenario 02 performance. This recovery demonstrates training data diversity substantially reduces worst-case boundary localization failures.

However, HD values remain slightly elevated compared to Scenario 01's 19.17–19.68 pixel range, indicating some challenging Qatar-COVID-19 images continue to produce larger boundary errors. The 1–8 pixel residual gap suggests domain shift effects persist even with enhanced training, though their magnitude decreased substantially.

Variance decreased dramatically. CNN-only's HD variance dropped from $\pm$ 16.39 pixels in Scenario 02 to $\pm$ 11.23 pixels here, while ViT-only improved from $\pm$ 13.76 pixels to $\pm$ 10.84 pixels. These reductions indicate more consistent worst-case performance.

HD95 Analysis (9.81–13.27 pixels)

HD95 improved impressively from Scenario 02's range of 16.87–26.69 pixels to 9.81–13.27 pixels. Hybrid Unweighted achieved best HD95 (9.81 $\pm$ 5.92 pixels), nearly matching Scenario 01's best result (8.23 pixels). This sub-10 pixel typical boundary error represents clinically acceptable segmentation quality for many applications.

Variance reductions are equally impressive. CNN-only's HD95 variance dropped from $\pm$ 12.14 pixels to $\pm$ 6.74 pixels, and ViT-only improved from $\pm$ 9.79 pixels to $\pm$ 7.00 pixels—approximately 40–50% reductions demonstrating diverse training data helps models handle varied presentations more consistently.

All three hybrid configurations achieved excellent HD95 values between 9.81–10.04 pixels, representing only 1–2 pixels degradation from Scenario 01 performance. This near-complete recovery indicates that for 95% of boundary points, enhanced training data successfully overcomes domain shift effects.

Summary of Training Diversity Effects

Enhanced training data diversity produced substantial improvements across all metrics and architectures. Recovery was most dramatic for IoU (6–7 points), sensitivity (3–5 points), and boundary metrics (halving HD95), while accuracy and specificity improved modestly since they were already near-ceiling in Scenario 02.

Variance reductions across all metrics indicate training diversity's primary benefit may be consistency rather than peak performance. Models trained on diverse data encounter fewer situations causing dramatic failures, resulting in more predictable clinical behavior.

Architectural rankings shifted: CNN-only recovered to match or exceed ViT-only performance, reversing Scenario 02's pattern where ViT demonstrated superior domain generalization. This suggests CNNs' local feature learning, while initially fragile to domain shift, scales more effectively with training diversity.

Surprisingly, Hybrid Unweighted achieved best overall performance across multiple metrics (Dice, IoU, Sensitivity, HD), despite adaptive loss weighting showing advantages in Scenario 01. This suggests that with sufficient training diversity, simpler unweighted loss combinations may actually generalize better than sophisticated adaptive strategies optimized for specific data characteristics.

Table 4.3

Performance on the Qatar-COVID-19 test set after enhanced training with MC+SHZ and additional normal COVID-19 images.

Model	**Acc (%)**	**Dice (%)**	**IoU (%)**	**Sens (%)**	**Spec (%)**	**Prec (%)**	**HD**	**HD95**
ViT Only	98.01 ± 0.86	95.96 ± 1.56	92.28 ± 2.83	95.58 ± 2.63	98.85 ± 0.42	96.39 ± 1.36	27.32 ± 10.84	13.27 ± 7.00
CNN Only	98.08 ± 1.02	96.06 ± 1.92	92.48 ± 3.49	94.66 ± 3.73	99.25 ± 0.20	97.59 ± 0.69	26.40 ± 11.23	12.22 ± 6.74
Hybrid (BCE+Dice Losses)	98.29 ± 0.86	96.52 ± 1.58	93.31 ± 2.90	95.21 ± 3.11	**99.36 ± 0.16**	**97.92 ± 0.56**	20.85 ± 9.45	10.04 ± 6.04
Hybrid (Adaptive Losses)	98.29 ± 0.84	96.51 ± 1.53	93.31 ± 2.81	95.55 ± 3.05	99.24 ± 0.17	97.56 ± 0.64	20.59 ± 9.42	9.90 ± 6.07
Hybrid (Unweighted Losses)	**98.38 ± 0.73**	**96.71 ± 1.32**	**93.67 ± 2.43**	**96.42 ± 2.35**	99.06 ± 0.32	97.05 ± 0.97	**20.28 ± 9.29**	**9.81 ± 5.92**

4.2.4 Comparative Analysis with State-of-the-Art Methods

Tables 4.4 and 4.5 present comprehensive comparisons of our proposed hybrid architecture against recent state-of-the-art methods. On the Montgomery + Shenzhen (M+S) dataset, our method achieved 96.65% DSC (Dice), 8.23 HD95, 93.52% IoU, 98.38% accuracy, 97.17% precision, and 96.13% recall, demonstrating competitive or superior performance across all metrics. Compared to recent transformer-based approaches, our method outperforms TransUNet (96.45% DSC) by 0.20 percentage points in Dice while achieving substantially better boundary precision with HD95 of 8.23 pixels versus TransUNet's 10.75 pixels—a 23% improvement in typical boundary error that translates to clinically meaningful precision gains. Against U-Net++ (95.83% DSC, 11.75 HD95) and Swin-Unet (95.71% DSC, 12.10 HD95), our hybrid architecture demonstrates clear advantages in both overlap metrics and boundary localization, with HD95 improvements of 30% and 32% respectively. The broader comparison across methods spanning 2020–2025 positions our work within the evolutionary trajectory of lung segmentation technology: while early approaches like VAE (2020) achieved 88.15% accuracy on M+S+C datasets, recent methods have converged toward high performance levels, with FAT-Net (2025) reaching 98.12% accuracy and 96.10% Dice on M+S datasets, and Swin U-Net (2024) achieving 97.80% accuracy and 95.71% Dice. Our method achieved 96.65% Dice on the M+S dataset, representing a 0.55 percentage point improvement over FAT-Net's 96.10%, achieved alongside superior boundary precision metrics (HD95 of 8.23 pixels) that most benchmark studies do not report. While these improvements appear modest in terms of overlap metrics, the substantial gains in boundary localization—23% better HD95 than TransUNet and 30–32% better than U-Net++ and Swin-Unet—demonstrate that our hybrid CNN-ViT architecture with cross-attention fusion provides meaningful clinical value through more precise segmentation boundaries, which are critical for accurate lung volume measurements and pathology detection near lung boundaries.

Table 4.4

Comparison with state-of-the-art methods. Dataset abbreviations: M=MC, S=SHZ, C=COVID19, P=Private, Q=QaTa-COV19. Results for FAT-Net are from Khorasani and Mofrad Khorasani and Mofrad (2025).

Research	**Method**	**Dataset**	**Acc (%)**	**Dice (%)**	**Year**
	VAE	M+S+C	88.15	-	2020
	U-Net	M+S	91.47	-	2022
	ConvMixer	M+S	97.52	-	2023
	VAE-AL-UNet	M+S+P	-	95.00	2023
	DECTNet	Q	97.50	87.99	2024
	Swin U-Net	M+S	97.80	95.71	2024
Khorasani and Mofrad (2025)	FAT-Net	M+S	98.12	96.10	2025
Ours	**Hybrid + Adaptive Losses**	**M+S**	**98.38**	**96.65**	**2025**
Ours	**Hybrid + Unweighted Losses**	**M+S+Q**	**98.38**	**96.71**	**2025**

Table 4.5

Performance comparison on the Shenzhen dataset. Benchmark results for U-Net, UNet++, TransUNet, and Swin-Unet are adapted from Yao et al. Yao et al. (2023).

Method	**DSC (%)**	**HD95**	**IoU (%)**	**Acc (%)**	**Prec (%)**	**Recall (%)**
U-Net	95.32	14.23	91.24	97.69	96.13	94.78
UNet++	95.83	11.75	92.15	97.95	97.31	94.62
TransUNet	96.45	10.75	93.25	98.16	97.36	95.72
Swin-Unet	95.71	12.10	91.88	97.80	96.81	94.79
Ours	**96.65**	**8.23**	**93.52**	**98.38**	**97.17**	**96.13**

4.2.5 Ablation Study Analysis

Tables 4.6 and 4.7 present ablation studies examining architectural components and loss function configurations across training scenarios.

Architectural Component Contributions

Under standard training (Scenario 1), both single-pathway models perform close to the full hybrid, with minimal Dice degradation. However, boundary precision reveals important differences: CNN-only shows substantial HD95 degradation while ViT-only maintains better boundary consistency, indicating ViT's global reasoning provides advantages even under favorable conditions.

Under cross-dataset generalization (Scenario 2), architectural importance shifts dramatically. CNN-only performance collapses with catastrophic boundary errors, while ViT-only degrades more gracefully. This demonstrates CNNs are substantially more brittle to domain shift than transformers, with CNN showing approximately five times larger Dice degradation and three times larger boundary error increases.

Enhanced training diversity (Scenario 3) partially reverses this pattern. CNN-only recovers to match ViT-only performance, suggesting sufficient training diversity addresses CNN brittleness. The full hybrid maintains consistent advantages across all scenarios, confirming architectural fusion provides value regardless of data conditions.

Loss Function Strategy Impact

Loss function effectiveness varies systematically with training conditions. In Scenario 1 with limited matched data, Adaptive Losses achieved best performance, demonstrating dynamic weighting helps stabilize training with constrained data. Under cross-dataset conditions (Scenario 2), this reversed: BCE+Dice generalized best, outperforming both Adaptive and Unweighted configurations. Adaptive weighting strategies may overfit to source domain properties, while simpler BCE+Dice generalizes more robustly under distributional shift.

With abundant diverse training (Scenario 3), Unweighted Losses dominated. BCE+Dice

showed degradation across Dice, IoU, and notably sensitivity, indicating the two-term loss configuration sacrifices recall for modest specificity and precision gains. Adaptive Losses performed similarly with smaller sensitivity loss but still degraded most metrics. Unweighted's superiority demonstrates that with diverse data, multiple complementary loss objectives with simple equal weighting outperform both limited loss diversity and sophisticated adaptive weighting.

Key Insights

The ablations reveal: (1) architectural fusion benefits scale with domain shift severity—minimal under matched distributions but substantial under cross-dataset conditions; (2) CNNs and ViTs exhibit complementary robustness profiles—CNNs excel with sufficient training diversity while ViTs demonstrate superior generalization under data scarcity or domain shift; (3) optimal loss strategy depends on data regime—adaptive weighting for limited matched data, simple BCE+Dice for domain shift, and multi-term fixed weighting for abundant diverse training.

Table 4.6

Performance comparison of the full hybrid model and its architectural ablations (CNN-Only, ViT-Only) across three experimental scenarios. The 'Full Hybrid' model serves as the baseline, with ΔDice and ΔHD95 metrics calculated relative to its performance in each scenario. Scenario 1 (Standard Training) tests on the MC+SHZ dataset. Scenarios 2 (Cross-Dataset Generalization) and 3 (Expanded Training) test on the unseen Qatar-COVID-19 dataset.

Configuration	Test Dataset	Training Data	Dice (%)	ΔDice vs Hybrid	HD95	ΔHD95 vs Hybrid
Scenario 1: Standard Training (640 images)						
Full Hybrid	**MC+SHZ**	**640**	$\mathbf{96.65 \pm 0.54}$	–	$\mathbf{8.23 \pm 1.23}$	–
w/o ViT (CNN-Only)	MC+SHZ	640	96.52 ± 0.57	-0.13%	9.65 ± 2.55	$+1.42$
w/o CNN (ViT-Only)	MC+SHZ	640	96.60 ± 0.65	-0.05%	8.55 ± 1.70	$+0.32$
Scenario 2: Cross-Dataset Generalization (640 training $\rightarrow$ COVID-19 testing)						
Full Hybrid	**COVID-19**	**640**	$\mathbf{94.56 \pm 2.15}$	–	$\mathbf{16.87 \pm 8.09}$	–
w/o ViT (CNN-Only)	COVID-19	640	92.27 ± 3.37	-2.29%	26.69 ± 12.14	$+9.82$
w/o CNN (ViT-Only)	COVID-19	640	94.14 ± 2.35	-0.42%	19.74 ± 9.79	$+2.87$
Scenario 3: Expanded Training (1,437 images $\rightarrow$ COVID-19 testing)						
Full Hybrid	**COVID-19**	**1,437**	$\mathbf{96.52 \pm 1.58}$	–	$\mathbf{10.04 \pm 6.04}$	–
w/o ViT (CNN-Only)	COVID-19	1,437	96.06 ± 1.92	-0.46%	12.22 ± 6.74	$+2.18$
w/o CNN (ViT-Only)	COVID-19	1,437	95.96 ± 1.56	-0.56%	13.27 ± 7.00	$+3.23$

Table 4.7

Performance comparison of different loss configurations for the hybrid model in Scenario 3. All Δ metrics are relative to the 'Unweighted Losses' baseline. Bold values indicate improvements over baseline; regular font indicates degradation. Note: For HD and HD95, negative Δ values represent improvements (lower is better).

Loss Configuration	**ΔAcc**	**ΔDice**	**ΔIoU**	**ΔSens**	**ΔSpec**	**ΔPrec**	**ΔHD†**	**ΔHD95†**
Unweighted Losses (Baseline)	–	–	–	–	–	–	–	–
BCE+Dice (Weighted)	-0.09%	-0.19%	-0.36%	-1.21%	$\mathbf{+0.30\%}$	$\mathbf{+0.87\%}$	$+0.57$	$+0.23$
Adaptive Losses	-0.09%	-0.20%	-0.36%	-0.87%	$\mathbf{+0.18\%}$	$\mathbf{+0.51\%}$	$+0.31$	$+0.09$

† For Hausdorff Distance metrics, negative values indicate improvement (shorter distance is better).

4.2.6 Model Complexity and Performance Trade-offs

Table 4.8 presents parameter count and performance trade-offs across architectural configurations. The ViT-Only model achieves competitive performance (95.96% Dice, 13.27 HD95) with only 6M parameters, representing the most parameter-efficient configuration. CNN-Only doubles this to 12M parameters, achieving marginally better Dice (96.06%) and improved boundary precision (12.22 HD95).

The hybrid architectures require 25M parameters—four times ViT-Only and twice CNN-Only. Despite this substantial overhead, Dice improvements appear modest: Hybrid Unweighted achieves 96.71% Dice, only 0.75 points above ViT-Only. However, boundary precision improvements are more substantial: hybrid models achieve HD95 of 9.81–10.04 pixels compared to 12.22–13.27 pixels for single-pathway models, representing 18–26% reductions in typical boundary error. This suggests the additional parameters primarily enhance boundary localization rather than overall segmentation overlap.

Within the hybrid architecture (constant 25M parameters), loss function variations show minimal impact: Unweighted achieves 96.71% Dice, Adaptive 96.51%, and BCE+Dice 96.52%. This indicates training strategy matters less than architectural choices for overall performance.

For deployment considerations, resource-constrained environments should consider ViT-Only (6M parameters), accepting slightly reduced boundary precision for 75% parameter reduction. Applications prioritizing boundary precision for clinical measurements should justify the hybrid architecture's 25M parameters, as the 18–26% boundary error reduction translates to meaningful improvements in lung volume estimation and pathology localization. The analysis demonstrates that optimal architectural choice depends critically on deployment context—screening versus diagnostic applications, computational constraints, and clinical accuracy requirements—rather than universal parameter efficiency metrics.

Table 4.8

Model Architecture Performance and Efficiency Trade-offs

Model	**Params**	**Dice (%)**	**HD95**
CNN-Only	12M	96.06 ± 1.92	12.22 ± 6.74
ViT-Only	**6M***	95.96 ± 1.56	13.27 ± 7.00
Hybrid (BCE+Dice Losses)	25M	96.52 ± 1.58	10.04 ± 6.04
Hybrid (Adaptive)	25M	96.51 ± 1.53	9.90 ± 6.07
Hybrid (Unweighted Losses)	25M	**96.71 ± 1.32**	**9.81 ± 5.92**

CHAPTER 5
CONCLUSION & DISCUSSION

Research Questions Conclusions

Parallel processing demonstrated superior performance, particularly under domain shift. Single-pathway models showed distinct weaknesses—CNNs degraded substantially on unfamiliar imaging protocols while ViTs produced imprecise boundaries. The parallel design preserved both pathways' strengths: CNN's local precision and ViT's global reasoning, without the compromises inherent in choosing one approach.

Bidirectional fusion enabled mutual refinement between two pathways. CNN features attended to ViT's global context, ensuring anatomically reasonable boundaries. ViT features attended to CNN's local details, achieving sharper localization. This mechanism is creating predictions satisfying both precision and consistency requirements.

The proposed architecture achieved competitive benchmark performance with distinct advantages in boundary precision and cross-dataset robustness. It maintained consistent performance across varied imaging conditions while offering computational efficiency suitable for clinical deployment.

Architectural Contribution

Proposed Architecture demonstrated that parallel CNN-ViT processing with bidirectional cross-attention provides an effective framework for medical image segmentation requiring both local precision and global consistency.

Methodological Contribution

Demonstrated that hybrid architectures and diverse training data address different generalization challenges—hybrids provide robustness to unfamiliar imaging conditions while diverse data improves all models. These represent complementary strategies rather than alternatives.

Limitations

The study concentrated on comparing different architectures and fusion methods. Other important factors like image preprocessing techniques, data augmentation strategies, or combining multiple models were not fully explored.

Performance was measured using mathematical accuracy scores (Dice, Hausdorff Distance). However, whether these improvements actually help doctors diagnose diseases faster or more accurately in real clinical settings remains unverified and needs testing with actual patients.

The study proved the concept works well in controlled conditions but needs broader testing with diverse real-world scenarios, exploration of complementary techniques, clinical validation with doctors, and deeper understanding of internal mechanisms before widespread hospital adoption.

Future Directions

Model Efficiency: Develop faster, lighter versions for practical deployment. Clinical Validation: Test with doctors across multiple hospitals to prove diagnostic value. Extended Use: Apply framework to other medical imaging tasks and non-medical domains. Interpretability: Visualize attention patterns to understand and improve the fusion mechanism.

Summary

This research developed a hybrid CNN-Vision Transformer architecture for chest X-ray lung segmentation, demonstrating that parallel feature processing with bidirectional cross-modal attention effectively balances local boundary precision and global anatomical coherence. The architecture achieved competitive performance with particular advantages in boundary accuracy and cross-dataset generalization, addressing key challenges in medical image segmentation. While limitations exist regarding evaluation scope and clinical validation, this work provides a practical foundation for clinical deployment and establishes principles applicable to broader medical imaging tasks requir-

ing multi-scale feature integration.

REFERENCES

Çallı, E., Sogancioglu, E., van Ginneken, B., van Leeuwen, K. G., and Murphy, K. (2021). Deep learning for chest x-ray analysis: A survey. *Medical Image Analysis*, 72:102125.

Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A. L., and Zhou, Y. (2021). Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*.

Chen, L.-C., Papandreou, G., Schroff, F., and Adam, H. (2017). Rethinking atrous convolution for semantic segmentation. *arXiv preprint arXiv:1706.05587*.

Çiçek, Ö., Abdulkadir, A., Lienkamp, S. S., Brox, T., and Ronneberger, O. (2016). 3d u-net: Learning dense volumetric segmentation from sparse annotation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 424–432. Springer.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

Fang, J., Lin, H., Chen, X., and Zeng, K. (2022). A hybrid network of cnn and transformer for lightweight image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1103–1112.

Gao, Y., Zhou, M., and Metaxas, D. N. (2021). Utnet: A hybrid transformer architecture for medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 61–71. Springer.

Hooper, S. (2023). *Label-Efficient Machine Learning For Medical Image Analysis*. PhD thesis, Standford University.

Hu, J., Shen, L., and Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 7132–7141.

Islam, J. and Zhang, Y. (2018). Towards robust lung segmentation in chest radiographs with deep learning. *arXiv preprint arXiv:1811.12638*.

Jadon, S. (2020). A survey of loss functions for semantic segmentation. *arXiv preprint arXiv:2006.14822*.

Jaeger, S., Candemir, S., Antani, S., Wáng, Y. X., Lu, P. X., and Thoma, G. (2014). Two

public chest X-ray datasets for computer-aided screening of pulmonary diseases. *Quantitative Imaging in Medicine and Surgery*, 4(6):475–477.

Kalinovsky, A. and Kovalev, V. (2016). Lung image segmentation using deep learning methods and convolutional neural networks. In *Proceedings of the International Conference*, pages 21–24, Minsk, Belarus. United Institute of Informatics Problems, Belarus National Academy of Sciences. Figure 2 shows Deep Encoder-Decoder Convolutional Neural Networks architecture.

Karimi, D. and Salcudean, S. E. (2019). Reducing the hausdorff distance in medical image segmentation with convolutional neural networks. *IEEE Transactions on Medical Imaging*, 39(2):499–513.

Khalil, E., Hussain, S., Umer, M., Nadeem, L., Alshardan, A., Alsubai, S., Mortajez Alharbi, A., and Ashraf, I. (2024). Automatic lung segmentation in chest x-ray images using sam with prompts from yolo. *IEEE Access*, 12:122816–122832.

Khorasani, M. S. and Mofrad, F. B. (2025). The effectiveness of FAT-Net in chest X-ray segmentation. *The Journal of Engineering*, 2025(1).

Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. (2017). Focal loss for dense object detection. *arXiv preprint arXiv:1708.02002*.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., and Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10012–10022.

Ma, J., Chen, J., Ng, M., Huang, R., Li, Y., Li, C., Yang, X., and Martel, A. L. (2021). Loss odyssey in medical image segmentation. *Medical Image Analysis*, 71:102035.

Oktay, O., Schlemper, J., Folgoc, L. L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N. Y., Kainz, B., et al. (2018). Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*.

Rahman, T., Khandakar, A., and et al. (2021). Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images. *Computers in Biology and Medicine*, 132:104319.

Rahman, T., Khandakar, A., Kadir, M. A., Islam, K. R., Islam, K. F., Mahbub, Z. B., Kadir, M. A., and Kashem, S. (2020). Reliable tuberculosis detection using chest x-ray with deep learning, segmentation and visualization. *IEEE Access*,

8:191586–191601.

Ronneberger, O., Fischer, P., and Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 234–241. Springer.

Shamshad, F., Khan, S., Zamir, S. W., Khan, M. H., Hayat, M., Khan, F. S., and Fu, H. (2023). Transformers in medical imaging: A survey. *Medical Image Analysis*, 88:102802.

Taghanaki, S. A., Zheng, Y., Zhou, S. K., Georgescu, B., Sharma, P., Xu, D., Comaniciu, D., and Hamarneh, G. (2019). Combo loss: Handling input and output imbalance in multi-organ segmentation. *Computerized Medical Imaging and Graphics*, 75:24–33.

Turk, O. and Kılıçaslan, S. (2025). Comparative analysis of u-net, v-net and seg-net architectures for lung segmentation. *PeerJ Computer Science*, 11:e2700.

Yao, W., Bai, J., Liao, W., Chen, Y., Liu, M., and Xie, Y. (2023). From CNN to Transformer: A Review of Medical Image Segmentation Models.

Yao, W., Bai, J., Liao, W., Chen, Y., Liu, M., and Xie, Y. (2024). From cnn to transformer: A review of medical image segmentation models. *arXiv preprint arXiv:2308.05305*.

Yuan, H., Hong, C., Tran, N. T. A., Xu, X., and Liu, N. (2024). Leveraging anatomical constraints with uncertainty for pneumothorax segmentation. *Health Care Science*, 3(6):456–474.

Yunusa, H., Qin, S., Chukkol, A. H. A., Yusuf, A. A., Bello, I., and Lawan, A. (2024). Exploring the synergies of hybrid cnns and vits architectures for computer vision: A survey. *arXiv preprint arXiv:2402.02941*. [cs.CV].

Zhang, Y., Liu, H., and Hu, Q. (2021). Transfuse: Fusing transformers and cnns for medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 14–24. Springer.

Zhou, Z., Rahman Siddiquee, M. M., Tajbakhsh, N., and Liang, J. (2018). Unet++: A nested u-net architecture for medical image segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, pages 3–11. Springer.